

Spent Ig } Abt in gate
Comp Ig }

$$\left(\frac{P_1}{P_2}\right)^{\frac{1}{\gamma}} = \left(\frac{T_1}{T_2}\right)^{\frac{1}{\gamma-1}}$$

$$\frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^{\gamma-1}$$

Info
when gas behaves as Ideal gas

pressure $\downarrow$ Temp $\uparrow$

* V. Temp

Perfect gas laws $P = P_{abs}$

$T = T_{abs}$

Boyle's law: $\left[P \propto \frac{1}{V} \right] (T=c)$ } Charles law $\left[V \propto T \right] P=c$
Gay-Lussac law $\left[P \propto T \right] V=c$

Ideal gas Eqn in terms

$$PV = mRT$$

$$P = \left(\frac{m}{V} \right) RT \Rightarrow \left[P = \rho RT \right] \quad \text{density}$$

Ideal gas Eqn in terms of no. of moles

$$PV = nRT$$

$$PV = \bar{n} \bar{R} T$$

dimensional homogeneity

$$P (\text{Pa}) \rightarrow R \left(\frac{\text{J}}{\text{kg} \cdot \text{K}} \right)$$

$$P (\text{kPa}) \rightarrow R \left(\frac{\text{kJ}}{\text{kg} \cdot \text{K}} \right)$$

$n = \text{No. of moles (kmol)}$

$\bar{R} = \text{Universal gas const} \left(\frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \right)$

Relationship (R) and ($\bar{R}$)

$$\left[R = \frac{\bar{R}}{M} \right]$$

$M = \text{Molecular weight}$

$$M_{O_2} = 32$$

$$M_{N_2} = 28$$

$$\frac{\text{kg}}{\text{kmol}}$$

$$\left\{ R = \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \right\} \left\{ \bar{R} \left(\frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \right) \right\}$$

gas const Universal gas const

$$\bar{R} = 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

Summary

Eqn in terms of
Eq. volume

$$PV = RT$$

Total Volume

$$PV = mRT$$

Density

$$P = \rho RT$$

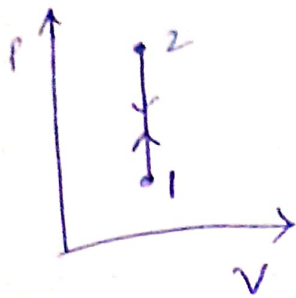
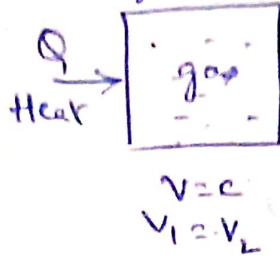
No. of moles

$$PV = \bar{n} \bar{R} T$$

01/12/24
Reversible

Const volume process

Ex: Rigid Tank

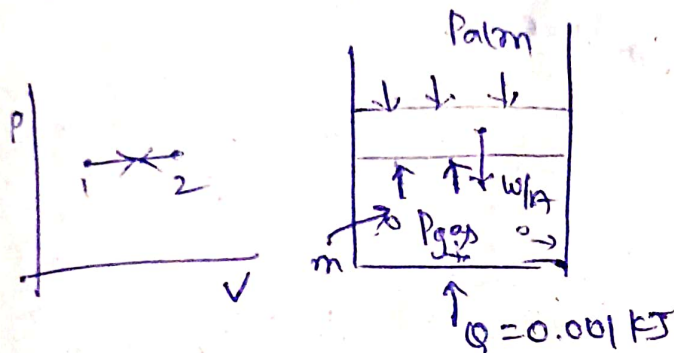


$$PV = nRT$$

$$\frac{PV}{T} = nR \Rightarrow \frac{PV}{T} = \text{const}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow \boxed{\frac{P_1}{T_1} = \frac{P_2}{T_2}}$$

Pressure process



$$P_{\text{up}} = P_{\text{down}}$$

$$P_{\text{gas}} = P_{\text{atm}} + \frac{W}{A}$$

$$P_{\text{gas}} = \text{const} + \text{const}$$

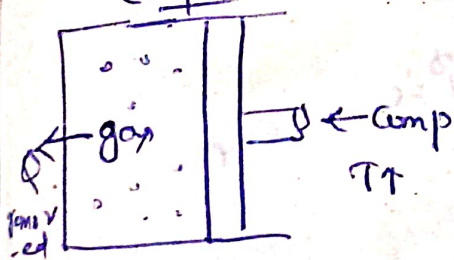
$$\boxed{P_{\text{gas}} = \text{const}}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

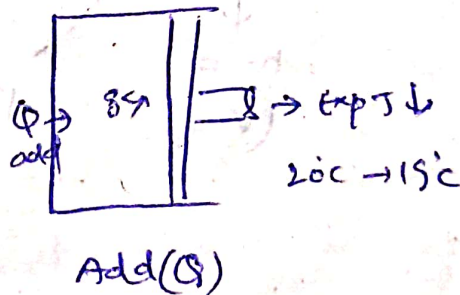
$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Ex: Slowly moving piston

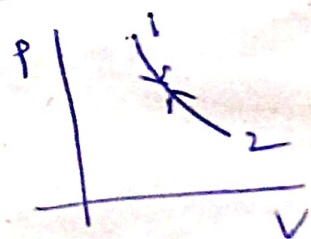
Const Temp process



Expansion



$$\boxed{P \cdot V = c}$$



Ex: adiabatic process

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$\gamma = 1.4$ gas $\rightarrow$ Air

General

$$\boxed{1 < n < \infty}$$

R.V. Imp.

$$n=0 \rightarrow P=c \rightarrow \text{Isobaric}$$

$$n=1 \rightarrow PV=c \rightarrow \text{Isothermal}$$

$$n=\gamma \rightarrow PV^\gamma=c \rightarrow \text{Adiabatic}$$

$$n=n \rightarrow PV^n=c \rightarrow \text{polytropic}$$

$$n=\infty \rightarrow V=c \rightarrow \text{Isochoric}$$

Expansion

$n > 0$ ($P=c$)

$n=1$ ($PV=c$)

$n=n$ ($PV^n=c$)

$n=\infty$ ($PV^\infty=c$)

$n=\infty$
($V=c$)

Comp

$n=n$

$n=2$

$n=\infty$ ($V=c$)

$n=1$

$n < 0$ ($P=c$)

$n=0$ ($P=c$)

$n=1$ $PV=c$

$PV^n=c$

$n=\infty$
($V=c$)

$PV^\infty=c$

Exp

02/11/24

Intensive and Extensive property

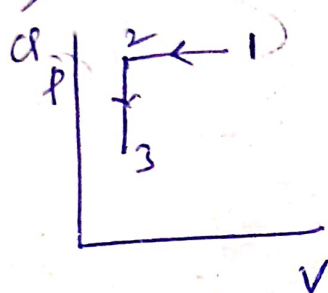
* Sp properties $\Rightarrow$ Intensive properties

Sp volume $\rho = \frac{V}{m} \rightarrow$ Volume
 $m \rightarrow$ mass

Rank

$$\text{Int pr} = \frac{\text{Ext pr}}{\text{Ext pr}}$$

04/11/24



$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{V_1}{T_1} = \frac{V_2}{\frac{3}{4} T_1} =$$

$$V_1 = \frac{4}{3} V_2 \Rightarrow \frac{V_1}{V_2} = \frac{4}{3} \Rightarrow \left(\frac{3}{4} = \frac{V_2}{V_1} \right)$$

Q.

No. of Molecules is Const in a Rigid tank and rigid tank means Const Volume

$$PV = nRT$$

$$dp V = n R dt$$

Quasi-static process $\rightarrow$ Extremely slow
 No friction

$$R = \frac{\bar{R}}{M_{H_2}}$$

molecular weight of $H_2 = 2$

Summary

mass (m) = $n \times M$

$n \rightarrow$ moles
 $M \rightarrow$ molecular mass
 $\rightarrow$ molar mass

$$m_{mix} = m_x + m_y$$

Summary

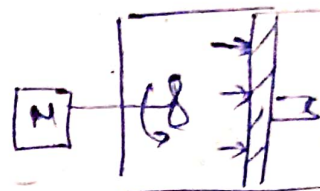
Work trf - Boundary phenomena, Transit form of energy, Transient (or) Transfer (Not Continuous)

closed sys work = $W = \int p dV$ valid for closed sys
Rev process

Electric work trf = $W_{el} = V \times I$

Riddle wheel work trf $W_{pad} = \text{Torque} \times \omega = \frac{2\pi NT}{60}$

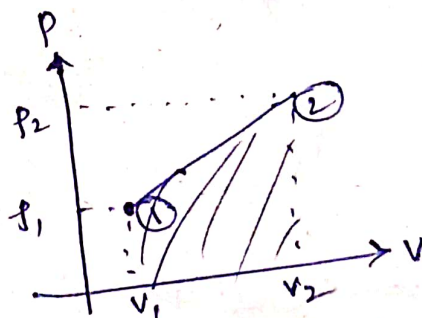
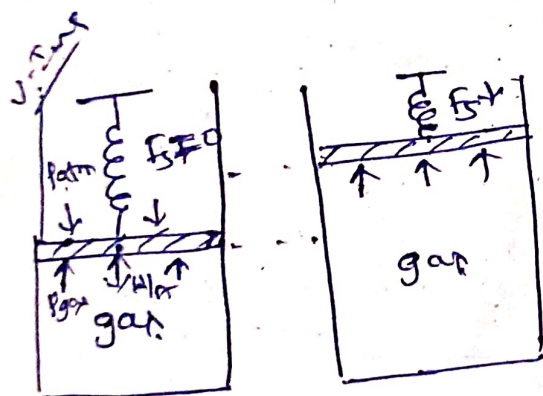
spring work trf $W_{spring} = \frac{kx^2}{2}$



$$W_{net} = W_{tot} = W_{mb} + W_{pad}$$

$$W_{net} = W_{mb} - W_{pad}$$

work given to sys



$$W = \frac{1}{2} (P_1 + P_2) (V_2 - V_1)$$

$\square + \triangle = \text{area}$

$$P_1 = P_{gas,1} = P_{atm} + \frac{W}{A} + F_s \rightarrow 0$$

$$P_2 = P_{gas,2} = (P_{atm} + \frac{W}{A}) + \frac{F_s}{A}$$

$$P_2 = P_1 + P_s \rightarrow \text{spring force}$$

$$P_1 = P_{atm} + \frac{W}{A}$$

problems

- 1. piston cylinder prob always $p=c$ and sym extremely slow frictionless

- Pressure reaches a value of absolute zero when molecular momentum of sym becomes zero

Heat & work

- Exact differential
- Path function
- Boundary phenomenon

$$R = \frac{\bar{R}}{M_m}$$

$$M_{N_2} = 28$$

$$M_{O_2} = 32$$

$$M_{CO_2} = 44$$

$$R_{mix} = \frac{m_{N_2} R_{N_2} + m_{O_2} R_{O_2}}{m_{N_2} + m_{O_2}}$$

Joule Thomson coeff (μ):

$$\mu = \left[\frac{dT}{dP} \right]_h$$

$$= \frac{+ve}{-ve} = -ve$$

$$\boxed{\mu < 0}$$

Heating

$$T \uparrow dT (+ve)$$

$$P \downarrow dP (-ve)$$

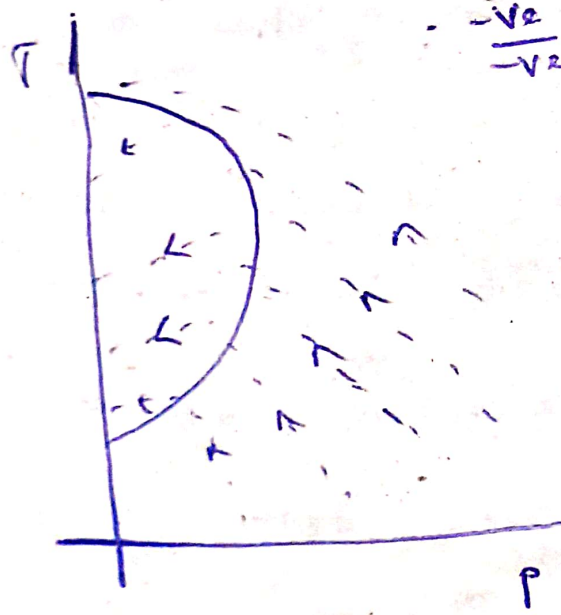
$$- \frac{-ve}{-ve} = +ve$$

$$\boxed{\mu > 0}$$

cooling

$$T \downarrow$$

$$T \downarrow dT (-ve)$$



→ when Real gas throttled $T \neq T_c$
 $T < T_c$

→ $\mu < 0$, $\mu > 0$, $\mu = 0$

→ slope of Isenthalpic line rep by μ
on p-T diagram

→ when Ideal gas is throttled $h = c$

$$\boxed{T_1 = T_2}$$

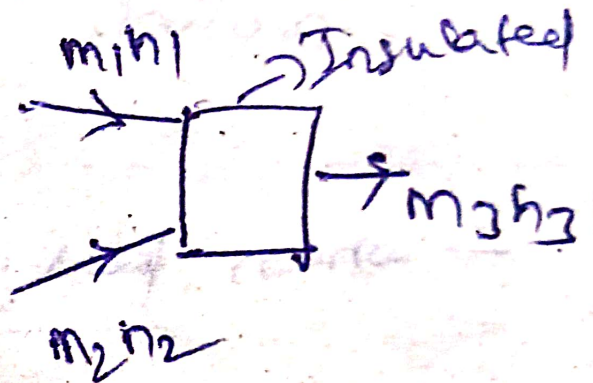
Overall Summary

$$\frac{\text{Energy In} - \text{Energy Out}}{\text{Energy}} = \frac{\text{Internal Energy} + Q_{\text{loss}}}{\text{Energy}}$$

$$m h_i = m u_f + Q$$

energy Bal Eqn

$$m_1 h_1 + m_2 h_2 = m_3 h_3$$



$$Q \neq \phi$$

$$m_1 = 0$$

$$\gamma_{air} = 1.4$$

$$T_2 = \gamma T_1$$

$$m \left(h_1 + \frac{V_1^2}{2000} + g z_1 \right) + Q = m \left(h_2 + \frac{V_2^2}{2000} + g z_2 \right) + W$$

Summary over all

$$\text{Total Energy of Sym (E)} = K.E + P.E + U$$

$$dE = d(K.E) + d(P.E) + dU$$

$$dE = dU$$

$$Q = dU + W \rightarrow \text{process}$$

$$Q = dH$$

$$Q = m c_p dT$$

$$C_p = 1.005$$

$$W = 0.718$$

$$R = 0.287$$

$$\gamma = 1.4$$

$$\frac{kJ}{kg \cdot K}$$

$$C_p - C_v = R$$

$$C_v = \frac{R}{\gamma - 1}, C_p = \gamma \left[\frac{R}{\gamma - 1} \right]$$

$$Q_{poly} = W_{poly} \left(\frac{\gamma - 1}{\gamma - 1} \right)$$

Throttling process

Irr process, friction involved, small opening so $p \uparrow$ & $v \uparrow$

$$\text{Open Sym Rev work by } W = - \int V dp$$

charging and discharging of tanks

$$T_2 = \gamma(T_1) - \text{Initial temp}$$

Suppose only rev (N) on given like 1000 rev
then $W_{pad} = (2\pi N) \times T$ KJ
 $= W \times T$

Ans

$$\text{paddle wheel } W.D_{pad} = \frac{2\pi N T}{60}$$

$$(ve \text{ sign})$$

need to take
b/c work given to sym

and substitute
in Energy Bal Eqn

$$Q = U + W$$

$$= U + (-W)$$

$$W = U$$

• always substitute work done in Energy Bal Eqn and then solve further
like $Q = dU + W \Rightarrow 0 = dU + (-100) \Rightarrow dU = 100 = m c_v dt$ then find what want

• When $T \uparrow$ air real gas behaviour on an ideal gas

• $m = \rho_1 A_1 V_1 = \rho_2 A_2 V_2 \Rightarrow \rho$ changes when flowing in nozzle

Summary overall

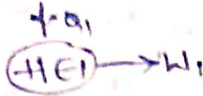


$$\eta_1 = 1 - \frac{T_2}{T_1}$$

or

$$T_2 = \sqrt{T_1 T_3}$$

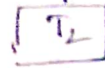
If $Q_1 = Q_2$



$$\eta_2 = 1 - \frac{T_3}{T_2}$$

$$T_2 = \frac{T_1 + T_3}{2}$$

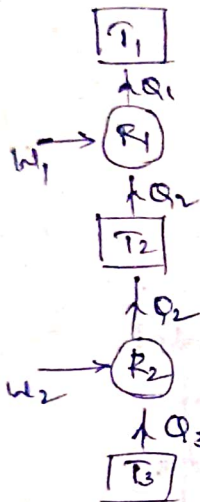
If W/output
 $W_1 = W_2$



$$\eta_{\text{over all}} = 1 - \frac{T_3}{T_1}$$

$$\eta_{\text{over all}} = \eta_1 + \eta_2 - \eta_1 \eta_2$$

Refrigeration



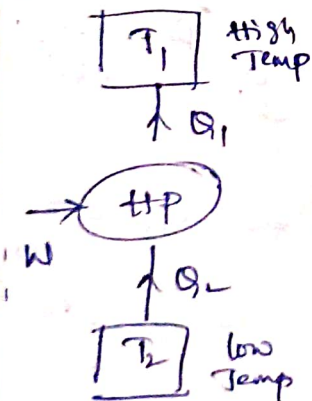
$$\text{COP}_1 = \frac{T_2}{T_1 - T_2}$$

$$\text{COP}_2 = \frac{T_3}{T_2 - T_3}$$

$$\text{COP}_{\text{over all}} = \frac{\text{COP}_1 \text{COP}_2}{1 + \text{COP}_1 \text{COP}_2}$$

$$\text{COP}_{\text{overall}} = \frac{T_3}{T_1 - T_3}$$

Cop. of heat pump



$$\text{COP} = \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2}$$

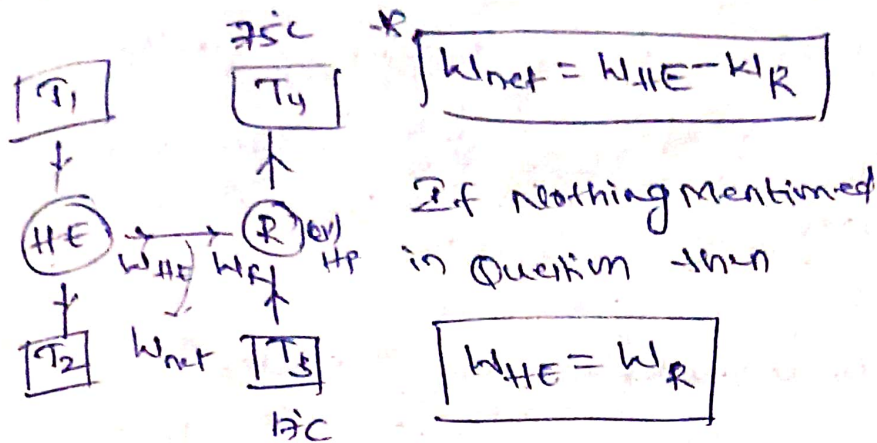
or simply

$$\text{COP}_{\text{HP}} = \text{COP}_R + 1$$

*

$$\text{COP}_{\text{HP}} = \frac{1}{\eta_{\text{HE}}}$$

Combined Heat Engine & Refrigerator



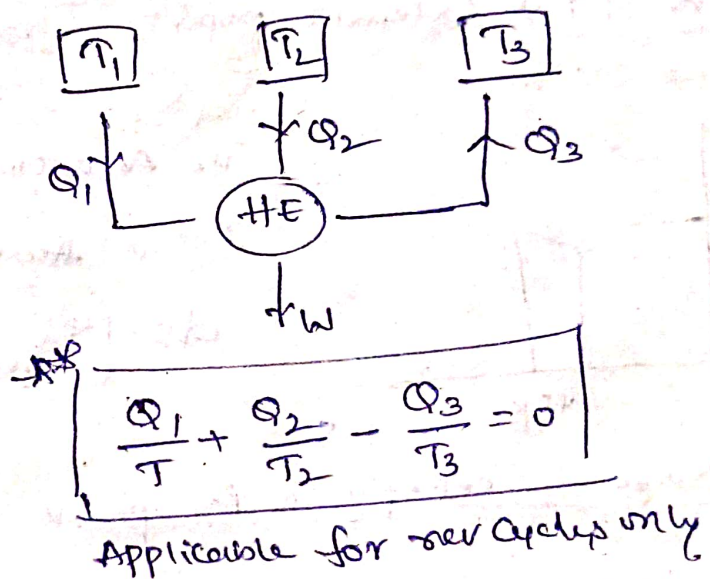
Clausius Eqn

$$\oint \frac{Q}{T} \leq 0$$

$$\oint \frac{Q}{T} = 0 \text{ Rev cycle}$$

$$\oint \frac{Q}{T} < 0 \text{ Irr. Rev cycle}$$

$$\oint \frac{Q}{T} > 0 \text{ Cycle Not possible}$$



Rev cycle
or
Carnot cycle

$$\eta_{HP} = \frac{T_1}{T_1 - T_2}$$

Irr cycle

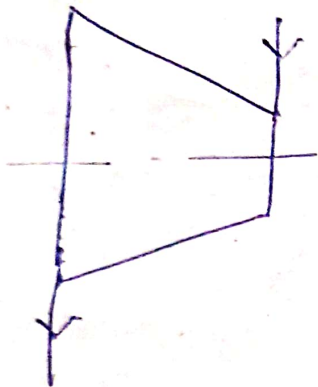
$$\eta = \frac{Q_1}{W}$$

$$\eta_{HP} = \frac{T_1}{T_1 - T_2}$$

not valid for Irr cycle

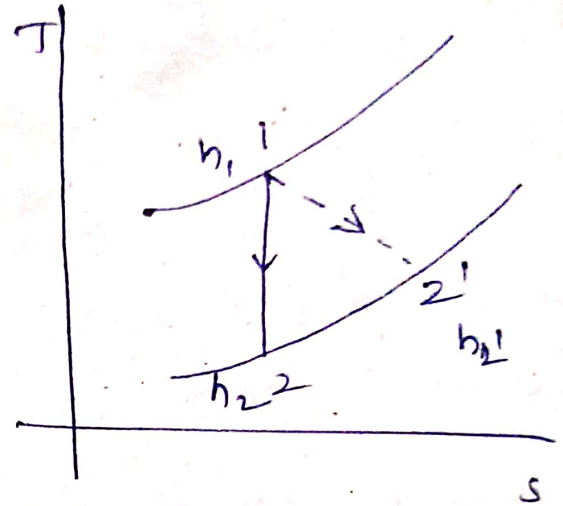
Isentropic efficiency (or) Adiabatic efficiency

Turbine



$$W_T = h_1 - h_2$$

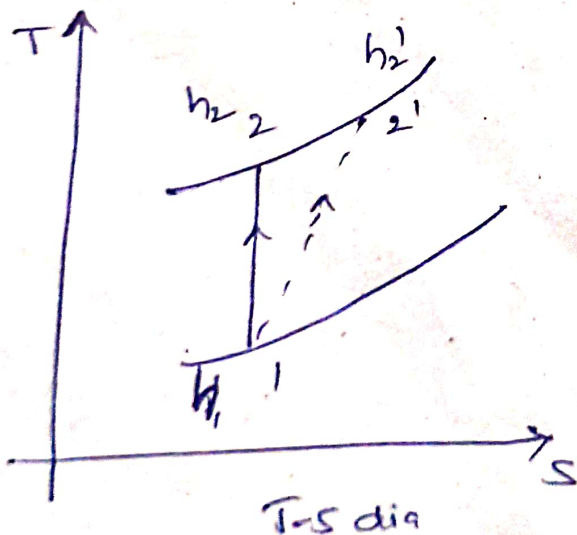
$$\eta_{isc-T} = \frac{W_{act}}{W_{ideal}} = \frac{h_1 - h_2'}{h_1 - h_2}$$



1-2 Rev Adia ($s=c$)

1-2' Irr Adia ($s \uparrow$)

Compressor



$$W_c = h_2 - h_1$$

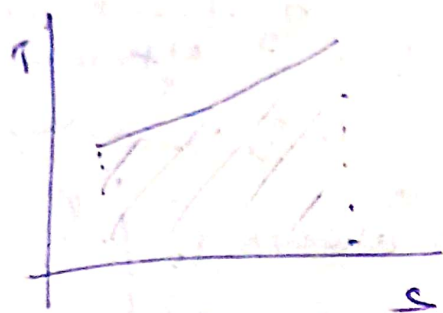
$$\eta_{isc-c} = \frac{W_{ideal}}{W_{act}}$$

$$\eta_{isc-c} = \frac{h_2 - h_1}{h_2' - h_1}$$

Summary

Entropy: Measure of disorderliness of molecules

$$S_{\text{gas}} > S_{\text{liq}} > S_{\text{solid}}$$

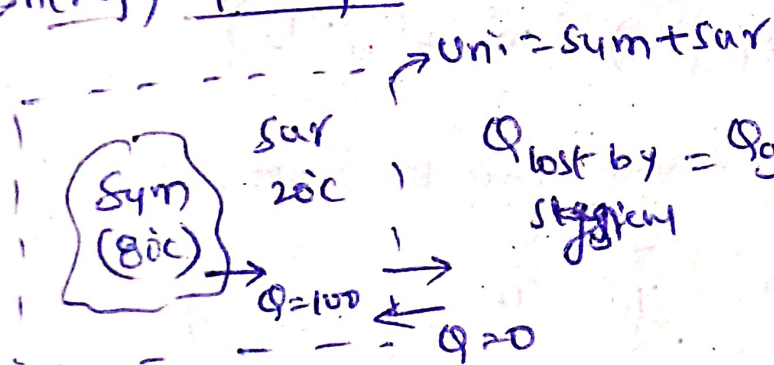


The Area under curve of TS diagram rep the Heat trf in rev process

In rev process ST , $ST \downarrow$ $S=c$ depends of Heat trf

Entropy principle

$$dS_{\text{uni}} = (dS_{\text{sym}} + dS_{\text{sur}}) \geq 0$$



$dS_{\text{uni}} = 0 \rightarrow$ Rev process
 $dS_{\text{uni}} > 0 \rightarrow$ Irr process
 $dS_{\text{uni}} < 0 \rightarrow$ Impossible process

Comp of First & Second law

First law for cycle

First law for process

$$Q = \Delta U + W$$

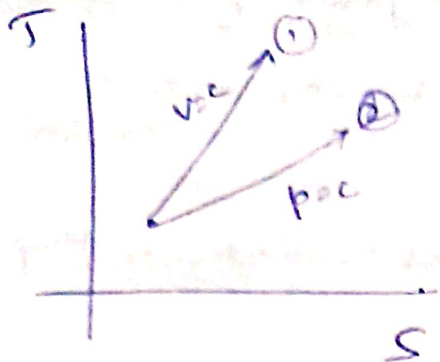
$$\oint Q = \oint W$$

Second law for cycle

Second law for process

$$\oint \frac{Q}{T} \leq 0$$

$$dS_{\text{uni}} \geq 0$$



$$\left(\frac{dT}{ds}\right)_v = \frac{T}{C_v} \quad \text{+ve}$$

$$\left(\frac{dT}{ds}\right)_p = \frac{T}{C_p} \quad \text{+ve}$$

$$\left(\frac{dT}{ds}\right)_v > \left(\frac{dT}{ds}\right)_p$$

$$(\text{slop})_{v=c} > (\text{slop})_{p=c}$$

Tds Equations

$$Tds = dh - vdp$$

$$Tds = du + pdv$$

change in entropy for solid and liq

$$\Delta S = mc \ln\left(\frac{T_2}{T_1}\right)$$

change in entropy due to mixing of liquids:

$$\Delta S_{\text{tot}} = \Delta S_{\text{sys}} = m_1 C_1 \ln\left(\frac{T_f}{T_1}\right) + m_2 C_2 \ln\left(\frac{T_f}{T_2}\right)$$

T_f = Final Temp after mixing

T_1, T_2 : Initial temp, liqs

$$\Delta S_{\text{uni}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \quad \boxed{\text{mixing process is always Irr}}$$

↓
If mixing process is Insulated

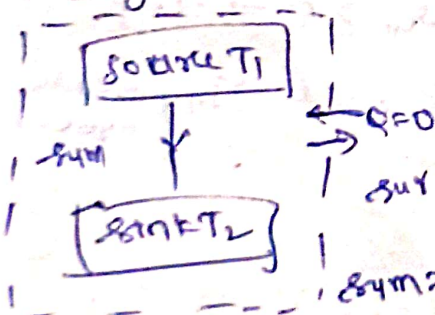
$$\boxed{T_f = \frac{m_1 C_1 T_1 + m_2 C_2 T_2}{m_1 C_1 + m_2 C_2}}$$

$$Q_{\text{lost}}(1) = Q_{\text{gain}}(2)$$

$$-m_1 C_1 (T_f - T_1) = m_2 C_2 (T_f - T_2)$$

Change in entropy thermal reservoirs

→ -ve b/c Q_{lost}



$$ds_{\text{sys}} = \left(\frac{-Q}{T_1}\right) + \frac{Q}{T_2}$$

$$ds_{\text{uni}} = ds_{\text{sys}} + ds_{\text{surr}} \quad (Q=0)_{\text{surr}}$$

$$\boxed{ds_{\text{uni}} = ds_{\text{sys}} = -\frac{Q}{T_1} + \frac{Q}{T_2}}$$

Per Adiabatic

$$ds = \left(\frac{Q}{T}\right)_{\text{rev}}$$

$$ds \geq 0$$

$$s=c$$

Isentropic process

Irr Adiabatic

$$ds = \left(\frac{Q}{T}\right)_{\text{Irr}} + S_{\text{gen}}$$

$$ds = S_{\text{gen}} (+ve) \quad ds > 0$$

(S↑) Always

Closed sym

Rev process

$$dS = \left(\frac{dQ}{T} \right)_{\text{rev}}$$

open sym

$$dS = \left(\frac{dQ}{T} \right)_{\text{rev}}$$

$$\Delta S = C_V \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{V_2}{V_1} \right)$$

$$\Delta S = C_P \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right)$$

$$\Delta S = C_V \ln \left(\frac{P_2}{P_1} \right) + C_P \ln \left(\frac{V_2}{V_1} \right)$$

Hot T_1
↓ Q
Cold T_2

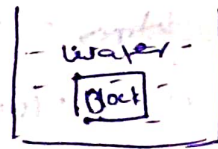
$$\Delta S_{\text{sym}} = \left(\frac{Q}{T_1} \right)_{\text{Hot}} + \left(\frac{Q}{T_2} \right)_{\text{Cold}}$$

$$\Delta S_{\text{Hot}} = -\frac{Q}{T_1}$$

$$\Delta S_{\text{Sur}} = \frac{Q}{T}$$

• metallic block immersed in water

Heat loss by Block = Heat gain



$$\Delta S = \int_{T_1}^{T_2} \frac{dQ}{T} = \int_{T_1}^{T_2} \frac{mc_p dT}{T} = mc_p \ln \left(\frac{T_2}{T_1} \right)$$

by water → By this find final temp

• $\Delta S_{\text{irr}} > \Delta S_{\text{rev}}$
→ Ambient air temp

• Irreversibility = $T_0 (dS)_{\text{uni}}$
(or)
loss in availability

$TdS = dU + pdV$ - To all processes and all working substance

$dQ = dU + pdV$ for rev process $\Rightarrow \delta W = pdV$ only for rev process

$$dS = \left(\frac{I^2 R}{T} \right)_{\text{power}}$$

$$T_f = \sqrt{T_H T_C}$$

Irr process

$$dS = \left(\frac{dQ}{T} \right)_{\text{Irr}} + S_{\text{gen}}$$

Irr process

$$dS = \left(\frac{dQ}{T} \right)_{\text{Irr}} + S_{\text{gen}}$$

$$dS = \left(\frac{dQ}{T} \right)_{\text{Irr}} + S_{\text{gen}}$$

Overall summary

for max work o/p
 $T_2 = T_4 = \sqrt{T_1 T_3}$

Otto cycle

$$\eta_{\text{Otto}} = 1 - \frac{1}{r^{\gamma-1}}$$

$$W_{\text{max}} = m c_v (\sqrt{T_3} - \sqrt{T_1})^2$$

Mean effective pressure

$$p_m = \frac{W_{\text{net}}}{V_s}$$

Diesel cycle

$$\text{Comp ratio } r_c = \frac{V_1}{V_2} = \frac{V_s + V_c}{V_c}$$

Cutoff ratio $r_{c,r}$

$$r = \frac{V_3}{V_2}$$

$$\eta_{\text{Diesel}} = 1 - \frac{1}{r^{\gamma-1}} \left[\frac{r^{\gamma} - 1}{\gamma(\gamma-1)} \right]$$

Comp ratio same then $\eta_{\text{Otto}} > \eta_{\text{Diesel}}$

$$\frac{V_2}{V_1} = \left(\frac{P_1}{P_2} \right)^{\frac{1}{\gamma}} = \left(\frac{T_1}{T_2} \right)^{\frac{1}{\gamma-1}}$$

Adiabatic relation

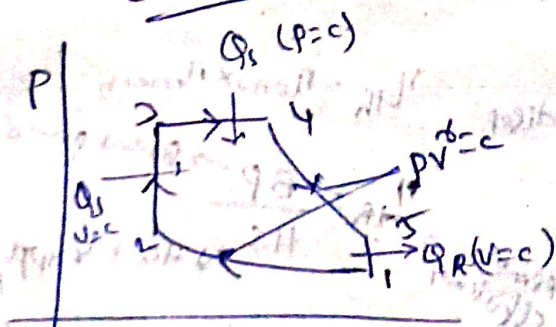
Dual cycle

$$\eta_{\text{dual}} = 1 - \frac{1}{r^{\gamma-1}} \left[\frac{r_c^{\gamma} \cdot r_p - 1}{(\gamma_p - 1) + \gamma r_p (\gamma_c - 1)} \right]$$

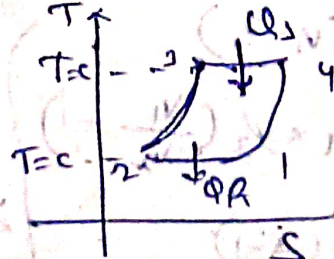
Generally $\eta_{\text{Otto}} < \eta_{\text{dual}} < \eta_{\text{Diesel}}$

If $r_c = 1$ $\eta_{\text{Otto}} = \eta_{\text{dual}}$
 If $r_p = 1$ $\eta_{\text{Diesel}} = \eta_{\text{dual}}$

Dual Cycle

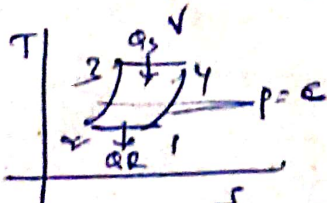


Stirling Cycle



$$\eta_{\text{Stirling}} = 1 - \frac{T_L}{T_H}$$

Erickson Cycle



$$\eta_{\text{Erickson}} = 1 - \frac{T_L}{T_H}$$

Back side one topic is written

1.9.19

$$V_{cut\ of\ k} = 10\% \cdot V_S$$

$$V_3 - V_2 = 0.1 (V_1 - V_2)$$

$$V_2 \left(\frac{V_3}{V_2} - 1 \right) = 0.1 \times V_2 \left(\frac{V_1}{V_2} - 1 \right)$$

$$g_c - 1 = 0.1 (g - 1)$$

$$g_c - 1 = 0.1 (17 - 1)$$

$$g_c = 2.6$$

∴ value

$$p = 1 + 0.1 (g - 1)$$

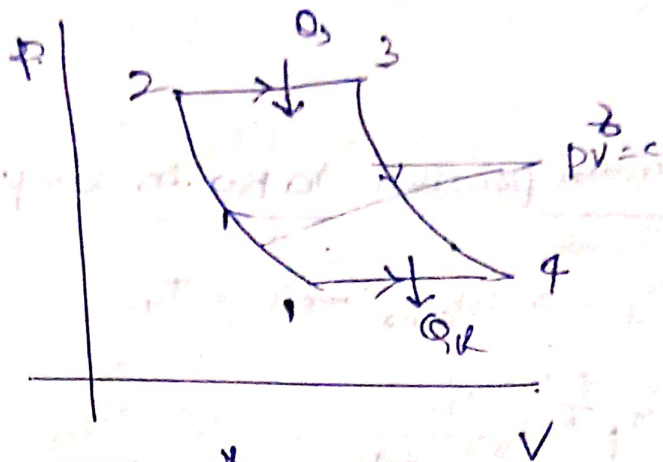
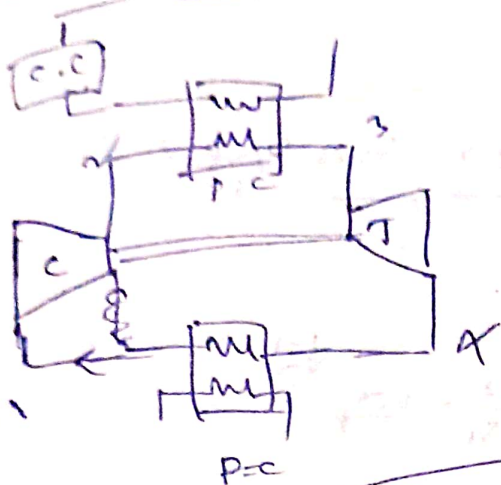
Cent 7/6

ndise

thermal
effi

Gas Turbines

closed cycle



Pressure ratio $\left(\frac{P_2}{P_1} = \frac{P_3}{P_4} \right)$

$\eta_{\text{Brayton}} = 1 - \frac{1}{\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}}$

S.F.E.E

$$\frac{W}{2} + g z_1 + h_1 + q = \frac{W}{2} + g z_2 + h_2 + W$$

$$h_1 + q = h_2 + W$$

Comp (1-2)

$$h_1 + q = h_2 - W_c$$

$$W_c = h_2 - h_1$$

(No work to be done)
H.E (2-3) heat to

$$h_2 + q = h_3 + W$$

$$q = h_3 - h_2$$

Turbine (3-4)

$$h_3 + q = h_4 + W_t$$

$$W_t = h_3 - h_4$$

$W=0$
H.E (4-1)

$$h_4 + q = h_1 + W$$

$$-q_R = h_1 - h_4$$

$$q_R = h_4 - h_1$$

$$\eta = \frac{W_{net}}{Q_s} = 1 - \frac{Q_R}{Q_s}$$

$$= \frac{W_t - W_c}{Q_s} = 1 - \frac{q_R}{q_s}$$

$$\eta = \frac{(h_3 - h_4) - (h_2 - h_1)}{h_3 - h_2} = 1 - \frac{h_4 - h_1}{h_3 - h_2}$$

Optimum pressure ratio in Simple Brayton Cycle:

$$\gamma_p \rightarrow W_{\max} \rightarrow T_2 = T_4$$

$$T_1 \gamma_p^{\frac{\gamma-1}{\gamma}} = \frac{T_3}{\gamma_p^{\frac{\gamma-1}{\gamma}}}$$

$$\frac{T_3}{T_1} = \gamma_p^{\frac{\gamma-1}{\gamma}} \times \gamma_p^{\frac{\gamma-1}{\gamma}}$$

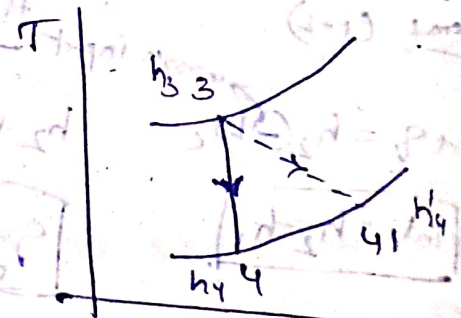
$$\boxed{(\gamma_p)_{\text{opt}} = \left(\frac{T_3}{T_1} \right)^{\frac{\gamma}{2(\gamma-1)}}$$

$$\frac{T_2}{T_1} = \gamma_p^{\frac{\gamma-1}{\gamma}}$$

$$\frac{T_3}{T_4} = \gamma_p^{\frac{\gamma-1}{\gamma}}$$

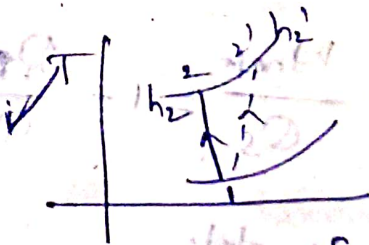
Isentropic η of Turbine

$$\eta_{\text{isentropic}} = \frac{W_{\text{act}}}{W_{\text{ideal}}} = \frac{h_3 - h_4}{h_3 - h_4'}$$

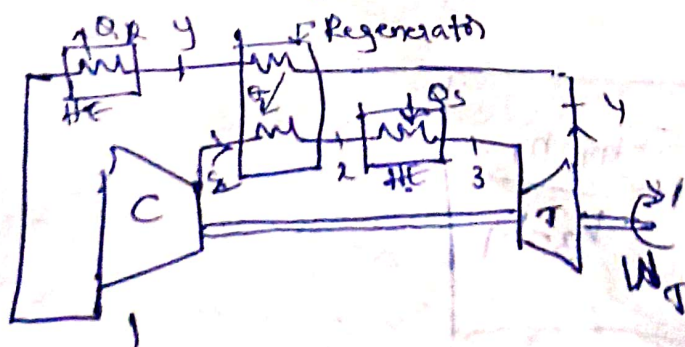


Isentropic η of Comp

$$\eta_{\text{isentropic}} = \frac{W_{\text{Ideal}}}{W_{\text{act}}} = \frac{h_2 - h_1}{h_2' - h_1}$$

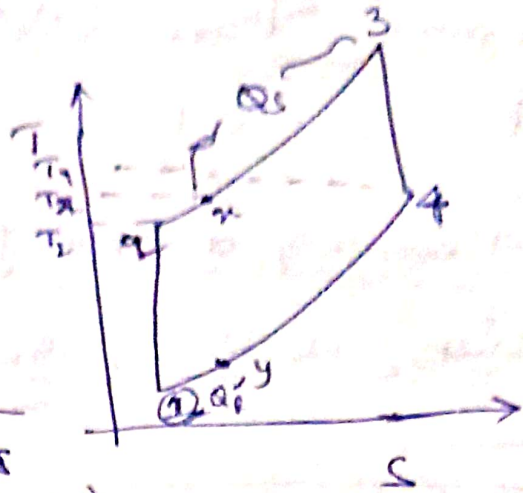
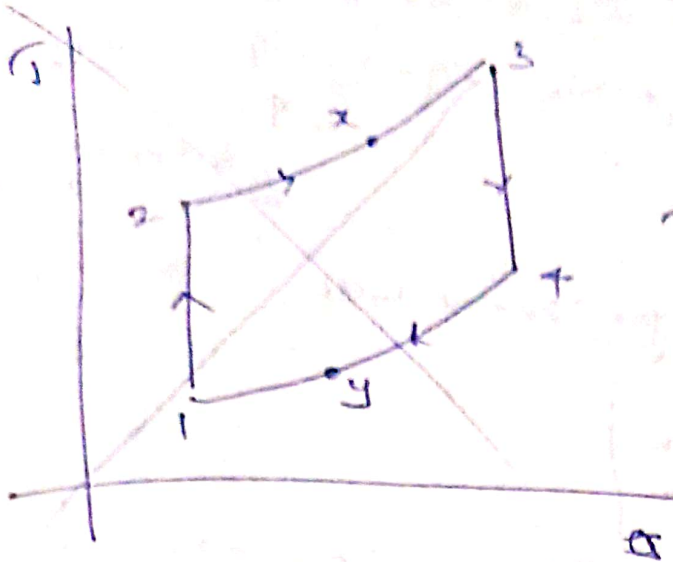


Regenerative gas Turbine



2-2' - Regeneration

3-3' - Regeneration



Effect due to regeneration (only)

$W_C = \text{same}$

$W_T = \text{same}$

$W_{net} = (W_T - W_C) \rightarrow \text{same}$

$Q_s \downarrow$

$$\uparrow \eta = \frac{W_{net}}{Q_s \downarrow}$$

Effectiveness of regeneration (ϵ)

$$\epsilon = \frac{\text{Actual heat gain in regeneration}}{\text{Ideal heat gain in regeneration}} = \frac{T_2 - T_2}{T_4 - T_2}$$

Reheat

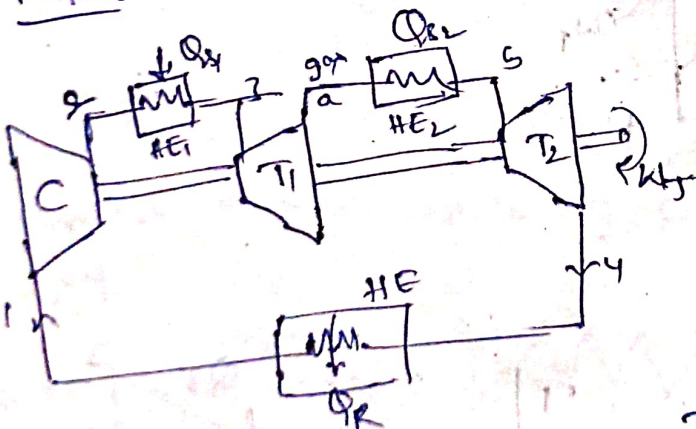
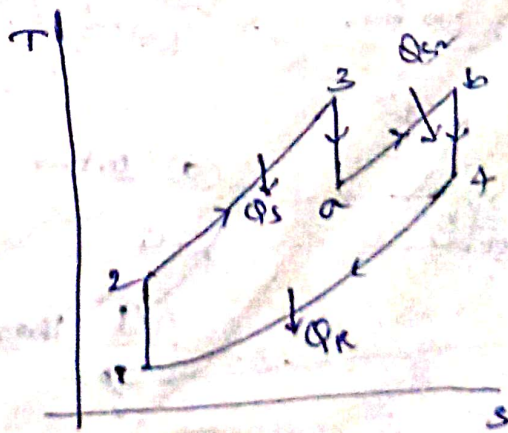


Diagram of heat



6354368240

Sadhu person in class of Gate

Admin of WhatsApp group

9100627211

Effects in Reheat gas turbine cycle:

- $W_c \Rightarrow$ same
- $W_T \uparrow$
- $Q_s \uparrow$
- $\dot{W}_{net} = (W_{T1} - W_c)$
- $\eta = \frac{W_{net}}{Q_s}$

η_H

Generally $\eta \uparrow$

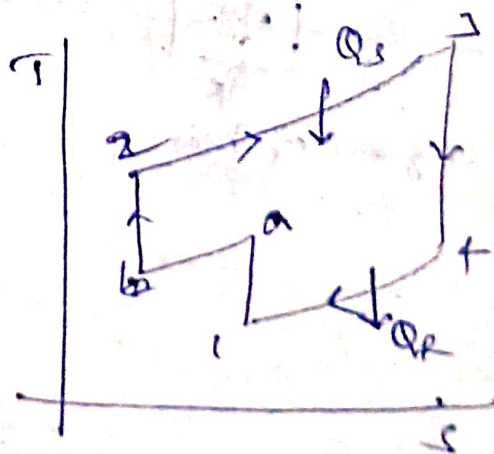
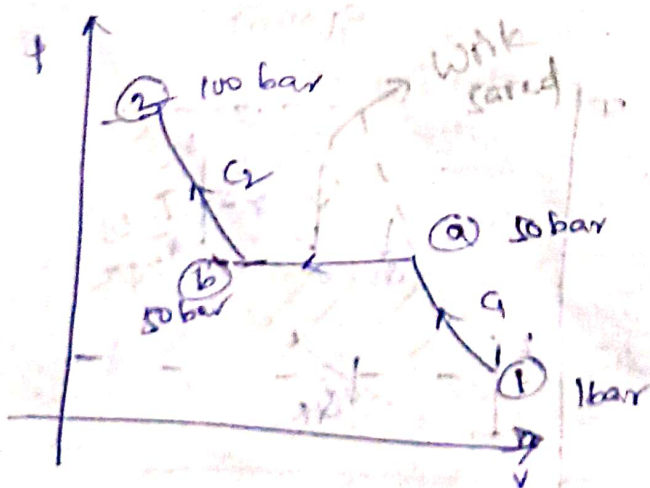
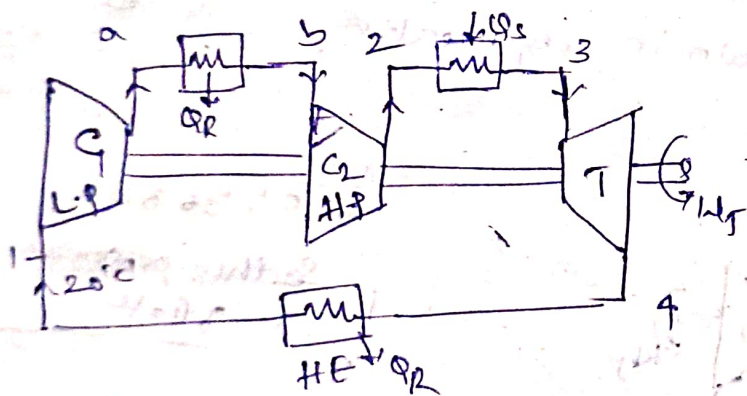
Thermal efficiency of Reheat gas turbine cycle:

$$\eta = \frac{W_{net}}{Q_s} = \frac{W_T - W_c}{Q_s} = 1 - \frac{Q_r}{Q_s}$$

$$W_T = W_{T1} + W_{T2} = (h_3 - h_a) + (h_b - h_4)$$

$$W_c = h_2 - h_1$$

Inter Cooling in gas turbine cycle:



Effect of Intercooling: (only)

$w_c \downarrow$
 $w_T \rightarrow \text{same}$
 $w_{net} = w_T - w_c \uparrow$
 $Q_s \uparrow$
 $\eta = \frac{w_{net}}{Q_s} \uparrow$

Thermal efficiency Intercooling
gas Turbine

$$\eta = \frac{w_{net}}{Q_s} = \frac{w_T - w_c}{Q_s} = 1 - \frac{Q_R}{Q_s}$$

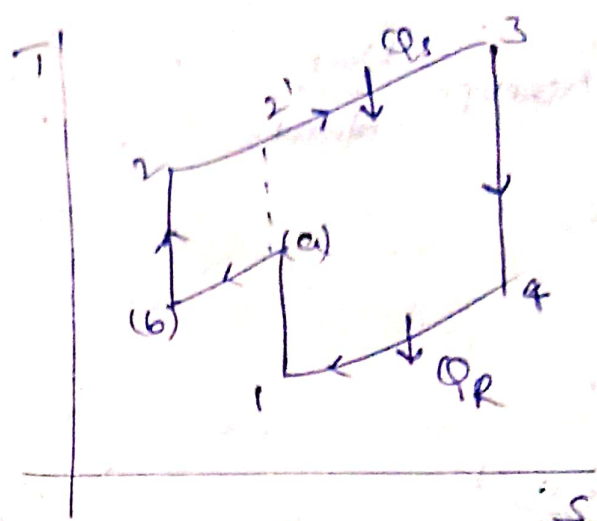
$$w_T = h_3 - h_4$$

$$w_c = w_{c1} + w_{c2} = (h_a - h_1) + (h_2 - h_b)$$

$$Q_s = h_3 - h_2$$

$$Q_R = h_4 - h_1$$

Back Work ratio (BWR) 1-



1-a-b-2-3-4-1 With Intercooling

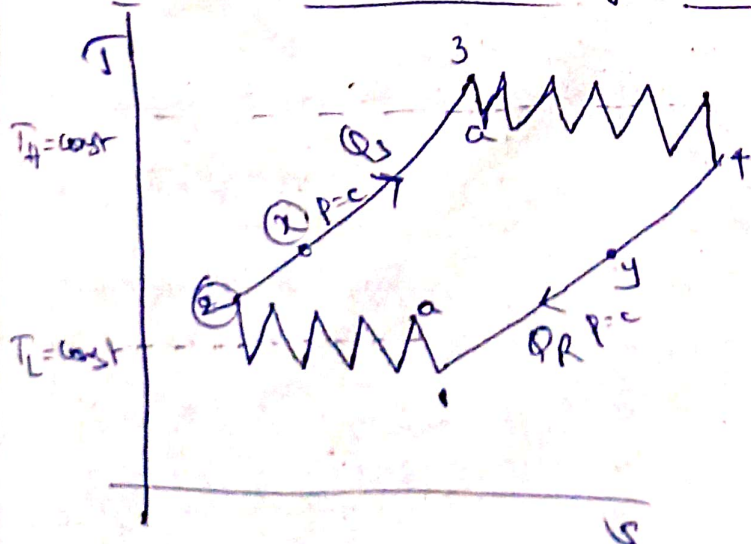
1-2'-3-4-1 Without Intercooling

Work ratio:

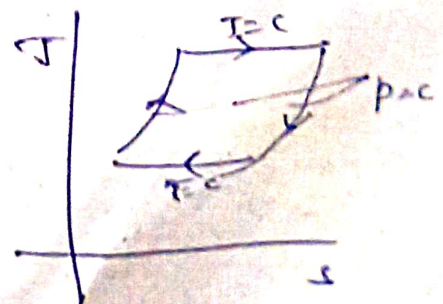
$$W.R = \frac{w_{net}}{w_{exp}}$$

$$W.R = \frac{w_T - w_c}{w_T}$$

Gas turbine with Intercooling, Reheating, Regeneration



It approaches Ericsson cycle



$$\eta_{Ericsson} = 1 - \frac{T_L}{T_H} = \eta_{Carnot}$$

prob 513

Ideal regeneration with
highest thermal efficiency

$$\eta_{\text{Ideal Reg GT}} = 1 - \frac{T_1}{T_3} (\gamma/p)^{\frac{\gamma-1}{\gamma}}$$

If many stages of compression
and expansion

$$\eta_{\text{Exicission}} = 1 - \frac{T_L}{T_H} = \eta_{\text{Carnot}}$$

Ideal gas - Eqn for γ or
adiabatic process

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

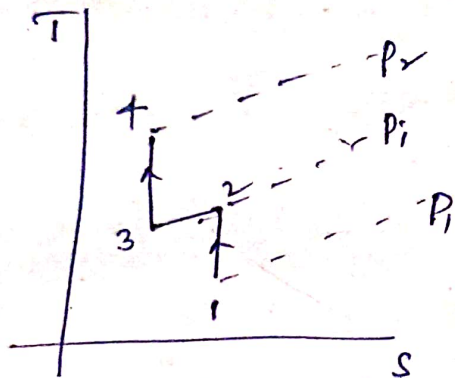
$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

$$(p)_\text{opt} = \left(\frac{T_\text{max}}{T_\text{min}} \right)^{\frac{\gamma}{\gamma-1}}$$

$$\eta_{\text{Isropic Turbine}} = \frac{h_3 - h_4'}{h_3 - h_4} = \frac{W_\text{act}}{W_\text{th}} = \frac{C_p (T_3 - T_4')}{C_p (T_3 - T_4)} = \frac{T_3 - T_4'}{T_3 - T_4}$$

Inter Cooler present



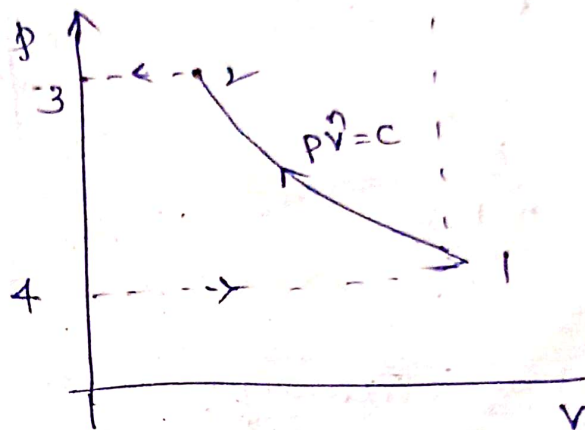
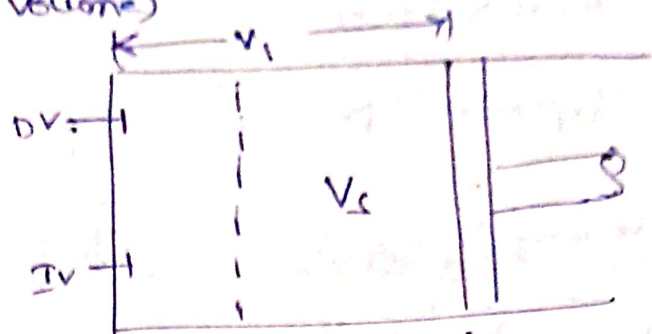
over all pressure ratio $\rho_0 = \left(\frac{P_2}{P_1} \right)^{\frac{1}{n}} \rightarrow \text{no. of stages}$

$$\eta = 1 - \frac{T_c}{T_h} = \eta_{\text{Carnot}}$$

Reciprocating Air Compressor

Single acting Reciprocating air compressor

Case 1: (without clearance volume)



$$W_{in} = \left(\frac{n}{n-1} \right) P_1 V_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

without clearance volume
($V_1 = V_s$)

Single acting Comp
Case 2 (with clearance)

4-1 Suction ($p=c$)

1-2 Compression ($pv^n=c$)

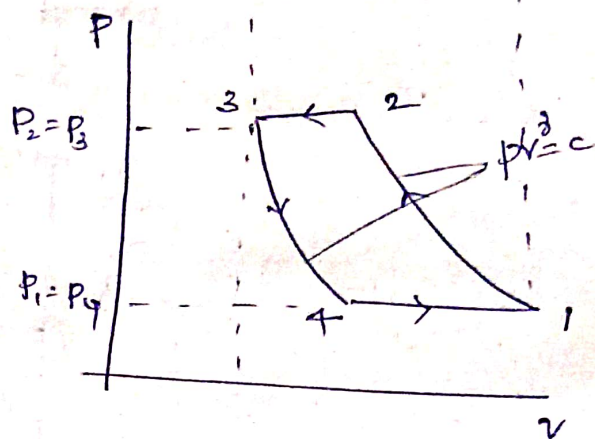
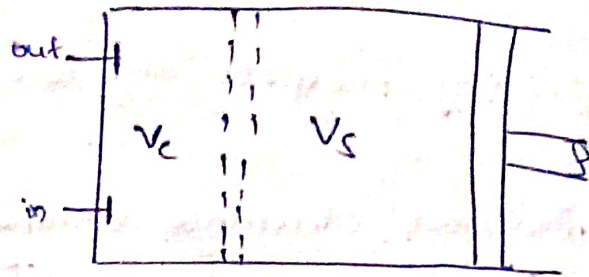
2-3 delivery ($p=c$)

3-4 Expansion ($pv^n=c$)

Theoretically $V_s = V_1 - V_3$

Actually $V_s = V_1 - V_4$

$$W_{in} = \frac{1}{n-1}$$



Volumetric Efficiency:

$$\eta_{vol} = \frac{(V_s)_{act}}{(V_s)_{th}} = \frac{V_1 - V_4}{V_1 - V_3}$$

$$= \frac{V_s + V_c - V_4}{V_s + V_c - V_c}$$

$$= \frac{V_s}{V_s} + \frac{V_c}{V_s} - \frac{V_4}{V_s} \times \frac{V_c}{V_c}$$

$$\eta_{vol} = 1 + c - c \left(\frac{V_4}{V_3} \right)$$

$$\eta_{vol} = 1 + c - c \left(\frac{P_2}{P_1} \right)^{\frac{1}{n}}$$

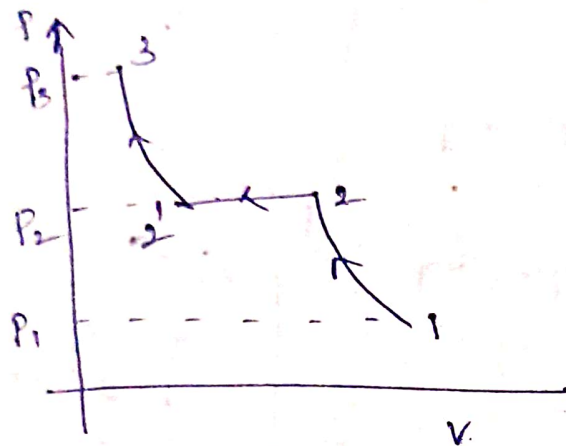
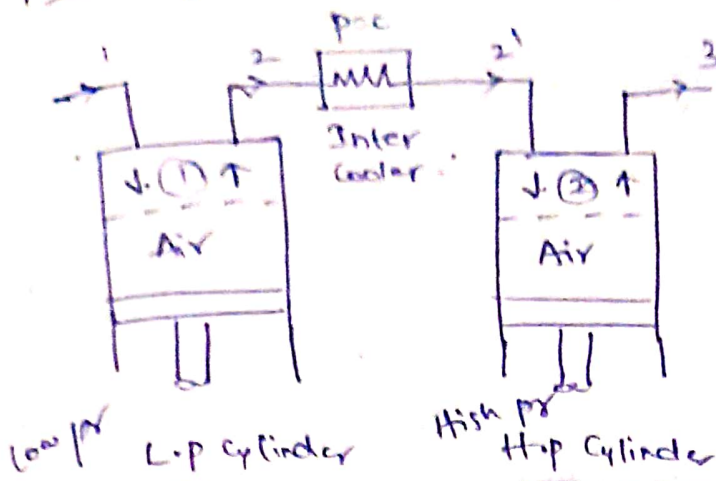
$$3-4 \quad pv^n = c$$

$$P_3 V_3$$

Multistage Compressor

2-2'

$T \downarrow$
 $V \downarrow$



Cond for perfect cooling.

$$T_1 = T_2'$$

$$mRT_1 = mRT_2'$$

$$P_1 V_1 = P_2' V_2' = mRT_1$$

Work Input to two stage reciprocating air compressor

$$(W_{in})_{\text{res}} = (W_{in})_1 + (W_{in})_2$$

$$= \left(\frac{n}{n-1}\right) P_1 V_1 \left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1 \right] + \left(\frac{n}{n-1}\right) P_2' V_2' \left[\left(\frac{P_3}{P_2'}\right)^{\frac{n-1}{n}} - 1 \right]$$

$$W_{in} = \left(\frac{n}{n-1}\right) mRT_1 \left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} + \left(\frac{P_3}{P_2}\right)^{\frac{n-1}{n}} - 2 \right]$$

$$P_2^2 = P_1 P_3$$

$$P_2 P_2 = P_1 \times P_3$$

$$\frac{P_2}{P_1} = \frac{P_3}{P_2}$$

$\propto P_1$ $\propto P_2$

$$P_2 = \sqrt{P_1 P_3}$$

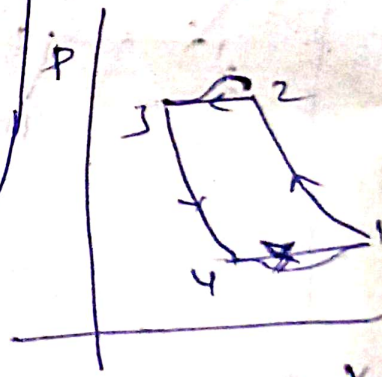
Note: for W_{in}

each stage pressure ratio should be same

Overall pressure ratio

$$\eta_{\text{overall}} =$$

$$\begin{aligned} (p/p)_{\text{each stage}} &= \sqrt{(p/p)_{\text{overall}}} && - 2 \text{ stage comp} \\ &= \left[(p/p)_{\text{overall}} \right]^{1/3} && - 3 \text{ stage comp} \\ (p/p)_{\text{each stage}} &= \left[(p/p)_{\text{overall}} \right]^{1/2} && - 2 \text{ stages comp} \end{aligned}$$



Power Input to Compressor

W'

Note

v. imp

$$W_{\text{double acting}} = 2 \times W_{\text{single acting.}}$$

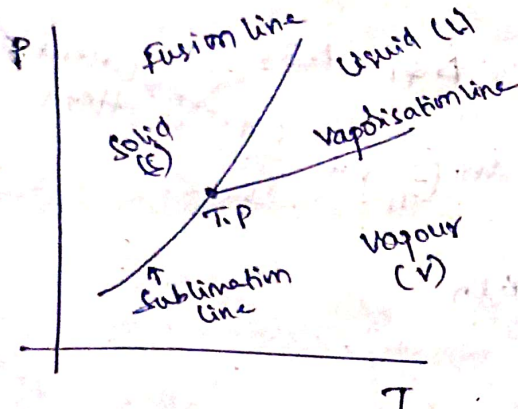
Dryness Fraction (or) Quality of Steam

Summary

Imp't

$$x = \frac{m_{\text{vap}}}{m_{\text{mix}}} = \frac{m_{\text{vap}}}{m_{\text{liq}} + m_{\text{vap}}}$$

P-T diagram for pure substance



Clausius - clapeyron Eqn

(or) Maxwell's Eqn

$$\left(\frac{\partial P}{\partial T}\right)_v = \left(\frac{\partial S}{\partial v}\right)_T$$

$$\frac{dp}{dT} = \frac{LH}{T_{\text{sat}}(v_g - v_f)}$$

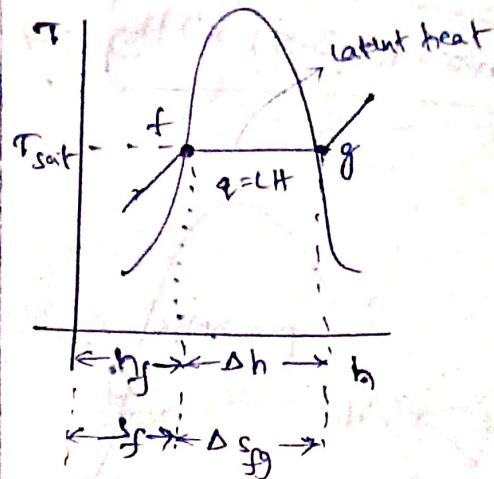
wet steam

$$v_{\text{wet}} = v_f + x v_{fg}$$

$$u_{\text{wet}} = u_f + x u_{fg}$$

$$s_{\text{wet}} = s_f + x s_{fg}$$

dry saturated steam



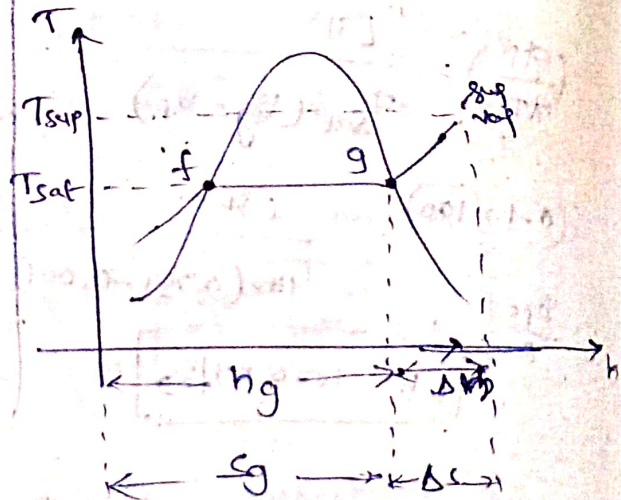
$$h_g = h_f + LH$$

$$h_f + h_{fg} = h_g$$

$$s_g = s_f + \frac{LH}{T_{\text{sat}}}$$

$$s_f + \Delta s_{fg} = s_g$$

superheated vapour



$$h_{\text{sup}} = h_g + \Delta h$$

$$h_{\text{sup}} = h_g + C_{p,\text{vap}} (T_{\text{sup}} - T_{\text{sat}})$$

$$s_{\text{sup}} = s_g + \Delta s$$

$$= s_g + C_{p,\text{vap}} \left(\frac{T_2}{T_1} \right) - \ln \left(\frac{T_2}{T_1} \right)$$

$$s_{\text{sup}} = s_g + C_{p,\text{vap}} \ln \left(\frac{T_{\text{sup}}}{T_{\text{sat}}} \right)$$

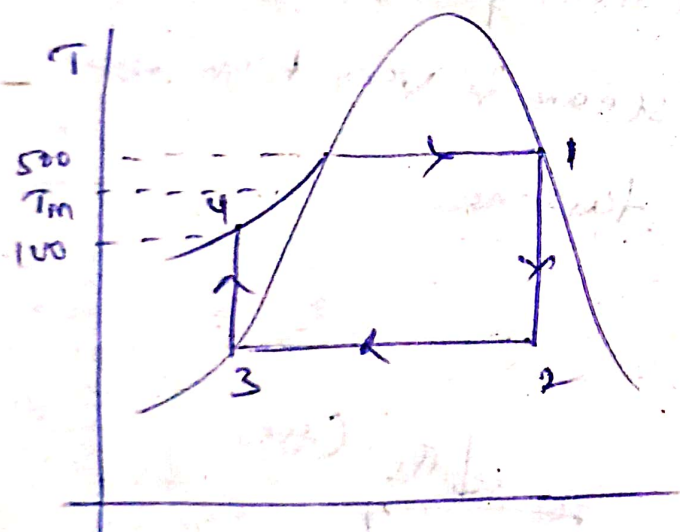
$$v_{\text{sup}} = x v$$

$$\eta = \frac{W_{net}}{Q_s} = 1 - \frac{Q_R}{Q_s}$$

$$= \frac{h_1 - h_2 - (h_4 - h_3)}{Q_s} = 1 - \frac{Q_R}{Q_s}$$

$$\eta_{Rankine} = \frac{(h_1 - h_2) - (h_4 - h_3)}{(h_1 - h_4)} = 1 - \frac{(h_2 - h_3)}{(h_1 - h_4)}$$

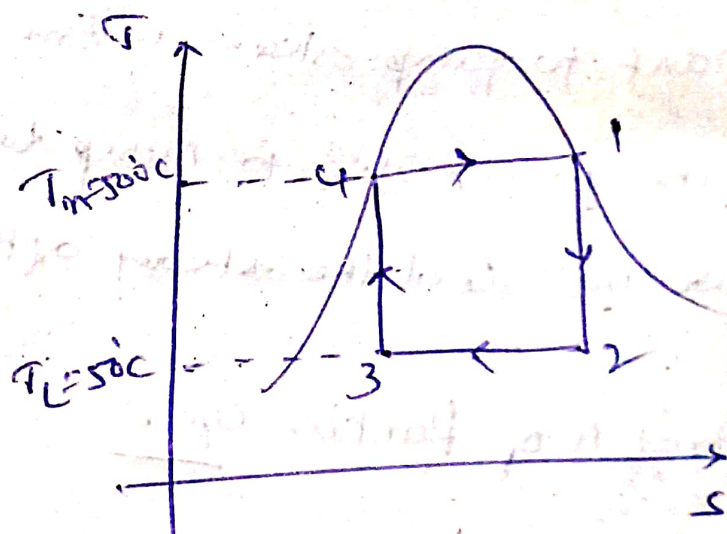
Mean temp in Rankine cycle



$$T_{mean} = T_{avg} = \frac{500 + 100}{2}$$

$$T_{avg} = 300^\circ C$$

Mean in Carnot cycle

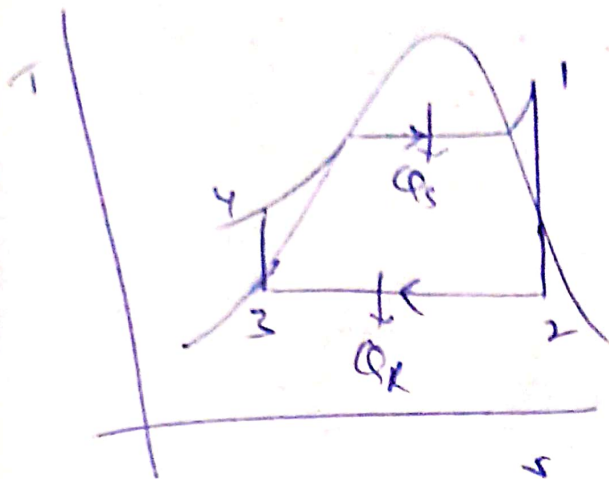


$$T_m = \frac{500 + 500}{2}$$

$$T_m = 500^\circ C$$

superheated Rankin Cycle

Effects of Superheating



Effect 1: $T_{mean} \uparrow$

Effect 2: $\eta \uparrow$

Effect 3: $W_T \uparrow$

Effect 4: W_P (same)

Effect 5: $\uparrow W_{net} = W_{T\uparrow} - W_P$

$$h_1 = h_{sup,1} = h_g + c_p (T_{sup} - T_{sat})$$

$$h_2 = h_f + x h_{fg}$$

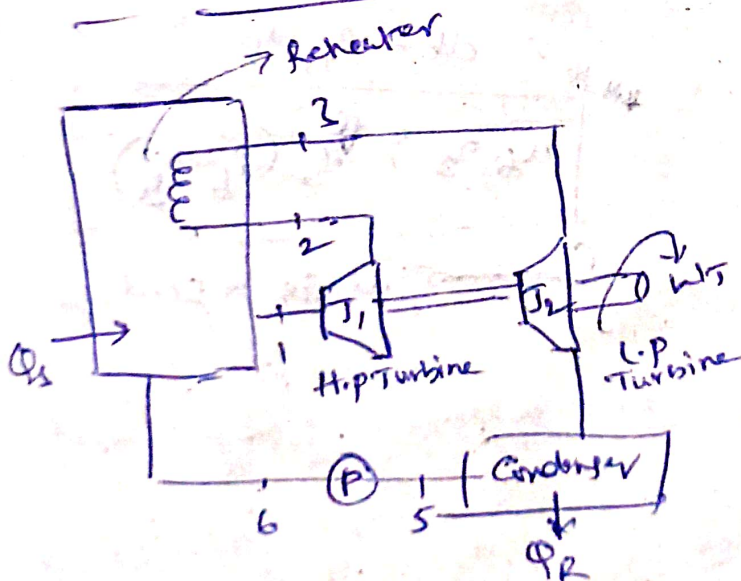
$$h_3 = h_{f,3}$$

$$\eta = \frac{W_T - W_P}{Q_S} = \frac{(h_1 - h_2) - (h_4 - h_3)}{(h_1 - h_4)}$$

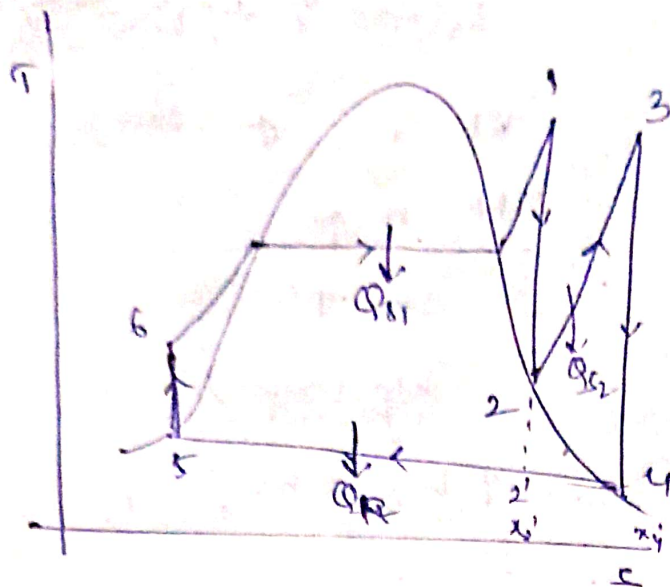
$$h_4 - h_3 = v_{f,3} (P_4 - P_3)$$

By using formula we need to find the h_4

Reheat Rankine cycle



Generally ($T_1 = T_3$)



Effect of Reheating

x_4 dryness fraction is more compared to x_2

^{Sup} The main purpose of reheating is to increase dryness fraction

- Mean $T_{mean} \uparrow$ so $\eta \uparrow$ (Generally in reheating T_m increases But not always?)
- $W_T \uparrow$
- W_P (same), $\uparrow W_{net} = \uparrow W_T - W_P$

The purpose of increasing the dryness fraction is to avoid the damage of blades by corrosion

$$\eta = \frac{W_{net}}{Q_s}$$

$$\eta = \frac{W_T - W_P}{Q_s} = 1 - \frac{Q_R}{Q_s}$$

$$W_T = (h_1 - h_2) + (h_3 - h_4)$$

$$W_P = h_6 - h_5$$

$$Q_s = (h_1 - h_6) + (h_3 - h_2)$$

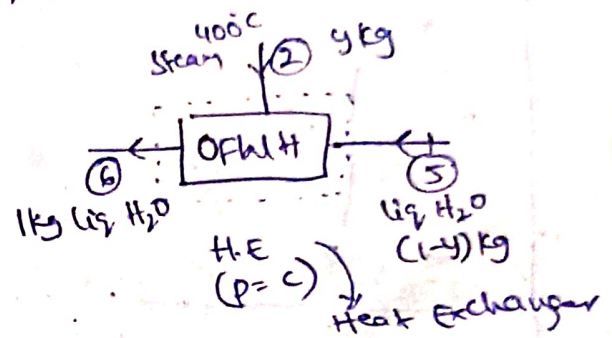
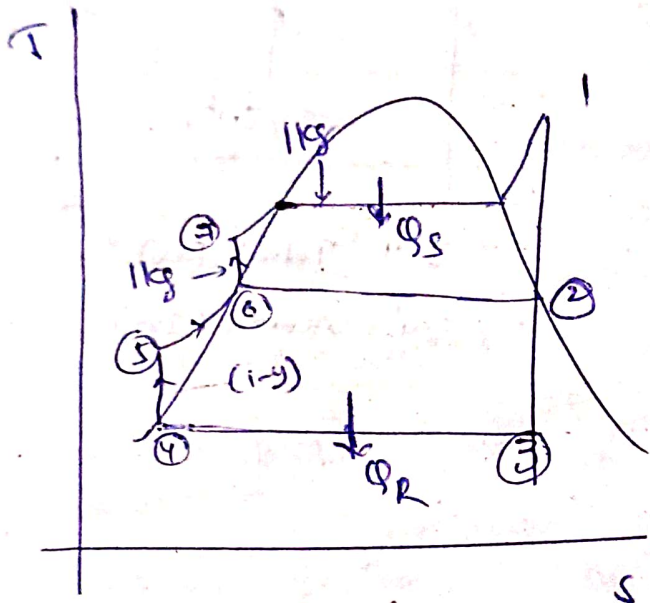
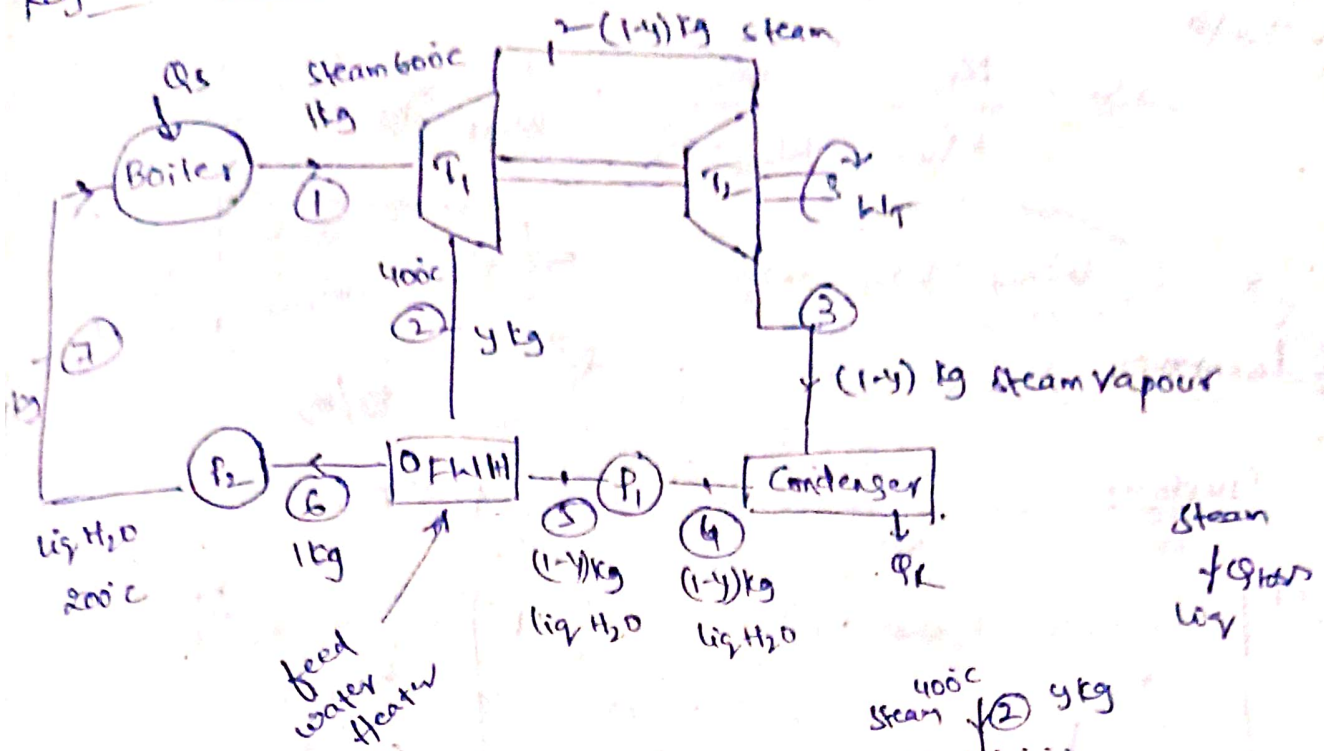
$$Q_R = (h_4 - h_5)$$

$$dh = v_{f5} dp$$

$$h_6 - h_5 = v_{f5} (p_6 - p_5)$$

Generalized eq to find h_6

Regenerative Rankin Cycle



$\eta_{\text{thermal}} \uparrow$
 $T_{\text{mean}} \uparrow$

Thermal η of Regenerative rankine cycle:

$$\eta = \frac{W_T - W_P}{Q_S}$$

Specific Steam Consumption (SSC):

Unit: $\frac{\text{kg}}{\text{kWh}}$

$$SSC = \frac{m_s}{W_{net}} = \frac{m_s}{W_{net} \times m_s} = \frac{1}{W_{net}} \left(\frac{\text{kJ}}{\text{kg}} \right) = \frac{1}{W_{net}} \left(\frac{\text{kJ}}{\frac{\text{kJ}}{3600}} \right)$$

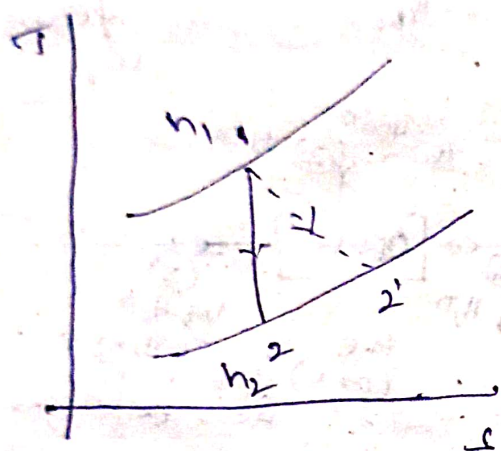
$$W_{net} = W_T - W_P$$

$$\Rightarrow SSC = \frac{3600}{W_{net}} \frac{\text{kg}}{\text{kWh}}$$

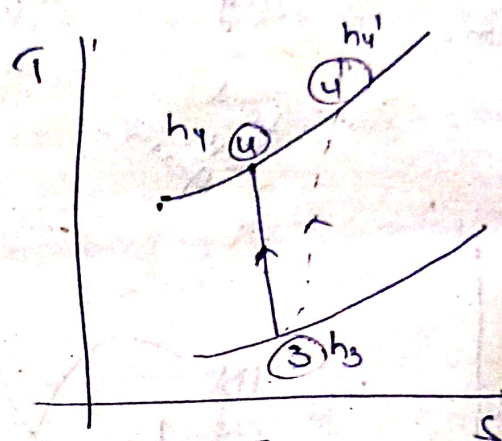
kJ/kg

Isentropic Efficiency:

Turbine

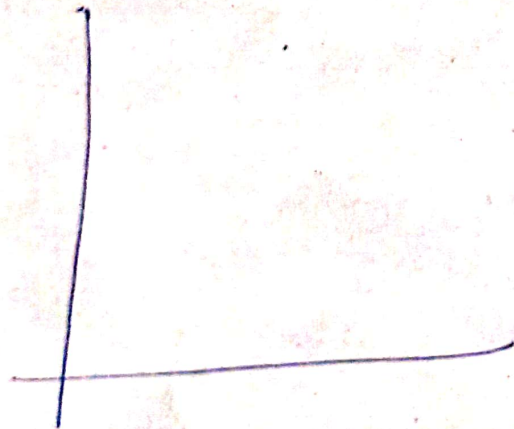


pump



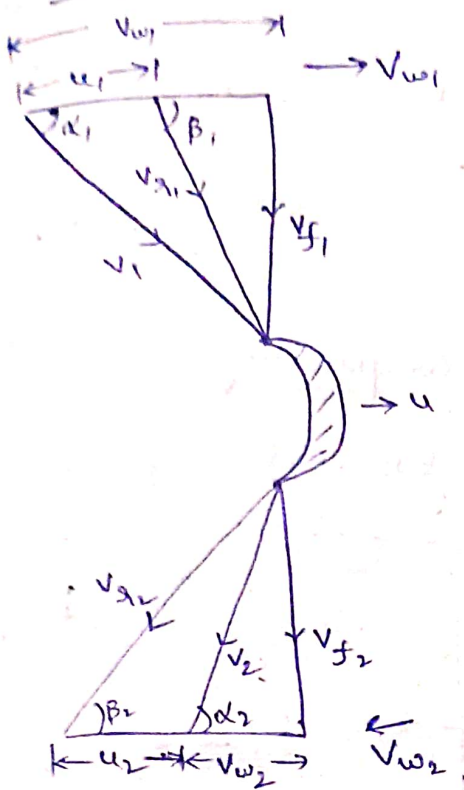
3-4 Ideal (Rev)

3-4' Actual (Irr)



Summary

Impulse Turbine velocity Triangles



$$W^2 = m u (V_{w1} + V_{w2}) \quad \text{Torque } (u_1 = u_2 = u)$$

Blade Spe.

Blade speed ratio (s)

$$s = \frac{u}{v_1}$$

Blade velocity coeff (or)

Blade friction factor (k)

$$k = \frac{v_{a2}}{v_{a1}}$$

Blade angle coeff (c)

$$c = \frac{\cos \beta_2}{\cos \beta_1}$$

Blade Efficiency (or)

Diagram Efficiency.

$$\eta = \frac{o/p}{I/p} = \frac{m u (V_{w1} + V_{w2})}{\frac{1}{2} m v_1^2}$$

$$\eta = \frac{2u(V_{w1} + V_{w2})}{v_1^2}$$

$$\eta_{max} = \frac{\cos^2 \alpha_1}{2} (1 + kc)$$

*** v.v. Imp ***

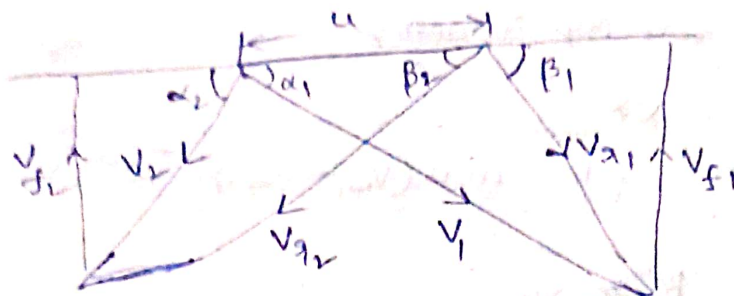
$$s = \frac{\cos \alpha_1}{2}$$

If $k=1, c=1$

$$\eta_{max} = \cos^2 \alpha_1$$

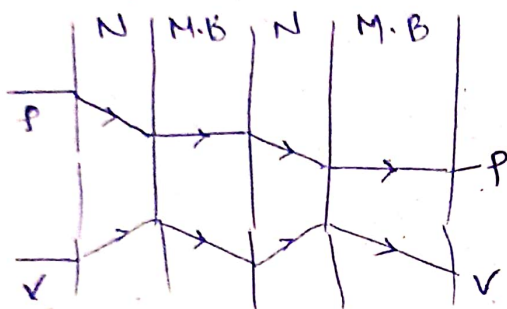
v.v. Imp

Combined Velocity diagram



Compounding

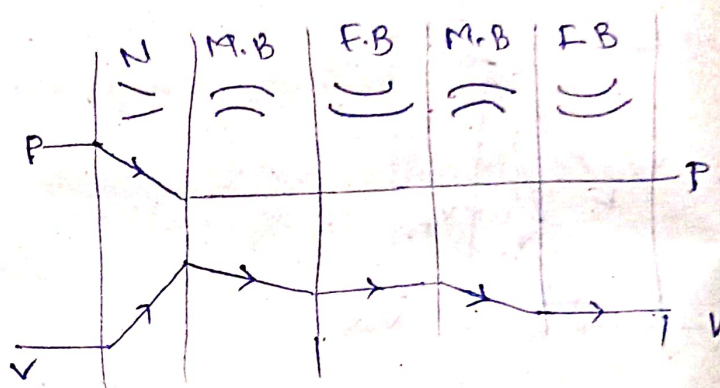
pressure Compounding



Pr ↓ in Stages

Ex: Rateau Turbine

Velocity Compounding



Velocity ↓ in stages, Fixed guide blades, Steam

$$V = C_{mgt}$$

Ex: Curtis Turbine

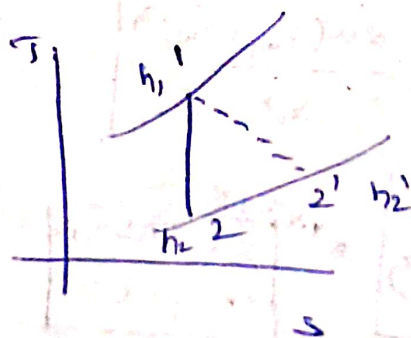
Reaction Turbine

$$\begin{bmatrix} \alpha_1 = \beta_2 \\ \alpha_2 = \beta_1 \end{bmatrix} \begin{bmatrix} V_1 = V_{1r} \\ V_2 = V_{2r} \end{bmatrix}$$

Nozzle Efficiency

$$\eta_{Lis-N} = \frac{\Delta h_{act}}{\Delta h_{isc}}$$

$$\eta_{Lis-N} = \frac{h_1 - h_2'}{h_1 - h_2}$$



Refrigeration

Refrigeration effect (or) Cooling effect

Refrigerant: It is a ^{or} fluid which extracts heat from refrigerated space

Cop of Refri

$$\text{COP}_R = \frac{\text{cooling effect}}{W_{\text{in}}}$$

Cop of HP

$$\text{COP}_H = \frac{\text{Heat eff}}{W_{\text{in}}}$$

Unit of Refrigeration

Ton of refrigeration (TR):

It is the amount of heat to be extracted from water at 0°C to ice at 0°C in 24hrs.

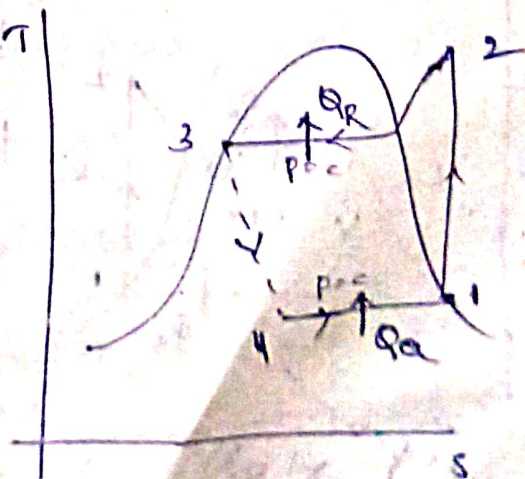
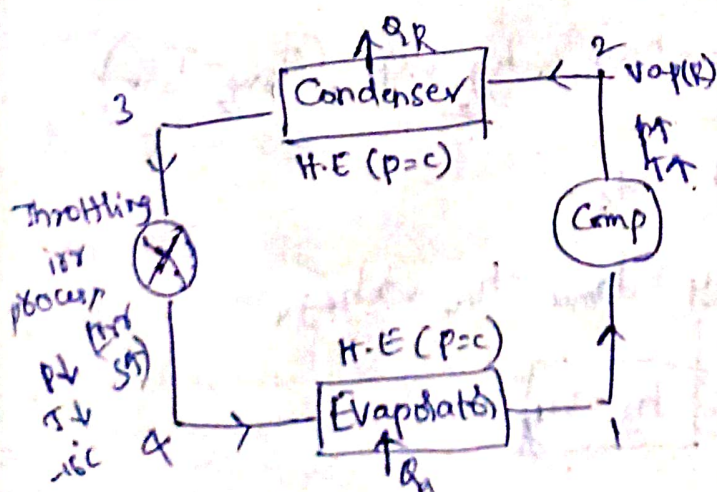
$$(Q_{\text{mfr}} = 907 \text{ kg}, \text{ LH} = 335 \frac{\text{KJ}}{\text{kg}})$$

$$\text{TR} = \frac{Q}{24 \text{ hrs}}$$

$$= \frac{\text{LH} \times m}{24 \times 3600} = \frac{335 \times 907}{24 \times 3600}$$

$$= \boxed{1 \text{ TR} = 3.516 \frac{\text{KJ}}{\text{s}}}$$

* Vapour Comp refrigeration cycle (VCR)



Cop of VCR cycle

$$\text{SFEE } \frac{V_1^2}{2} + gz_1 + h_1 + q = \frac{V_2^2}{2} + gz_2 + h_2 + W$$

k.E, p.E Negligible $\boxed{h_1 + q = h_2 + W}$

Compressor (1-2)

$$h_1 + q_v^0 = h_2 - W_c$$

$$\boxed{W_c = h_2 - h_1}$$

Throttle valve (3-4)

$$h_3 + q_v^0 = h_4 + W^0$$

$$\boxed{h_3 = h_4}$$

Condenser (2-3)

$$h_2 - q_R = h_3 + W^0$$

$$\boxed{q_R = h_2 - h_3}$$

Evaporator (4-1)

$$h_4 + q_a = h_1 + W^0$$

$$\boxed{q_a = h_1 - h_4}$$

Cop of VCR cycle

$$\text{COP}_{\text{VCR}} = \frac{\text{Cooling effect}}{W_c} = \frac{q_a}{W_c}$$

$$\boxed{\text{COP}_{\text{VCR}} = \frac{h_1 - h_4}{h_2 - h_1} = \frac{h_1 - h_3}{h_2 - h_1}}$$

Enthalpies of various points

$$S_2 = S_{\text{sup}} = S_{g_1} + C_{p,\text{vap}} \ln \left(\frac{T_{\text{sup}}}{T_{g_1}} \right)$$

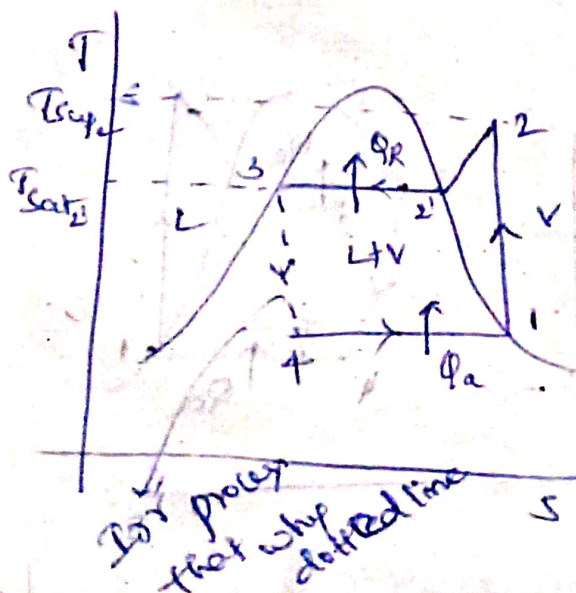
$$h_1 = h_{g_1}$$

$$h_2 = h_{\text{sup}_2} = h_{g_2} + C_{p,\text{vap}} (T_{\text{sup}_2} - T_{\text{sat}_2})$$

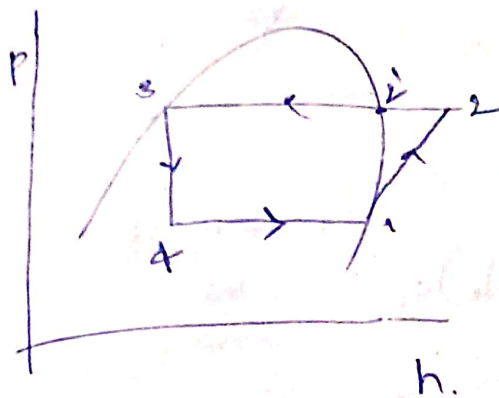
$$h_3 = h_{f_3}$$

$$h_4 = h_{\text{mix}} = h_{f_4} + x h_{fg_4}$$

$$\boxed{h_3 = h_4}$$



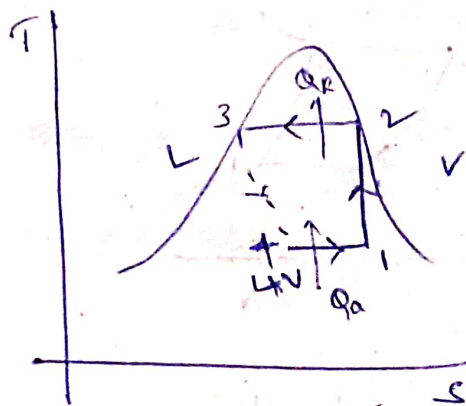
p-h diagram of Ideal VCR or stand VCR cycle:



$$s_2 = s_{up} = s_{g2} + c_{p, \text{vap}} \ln \left(\frac{T_{\text{sup}2}}{T_{\text{sat}2}} \right)$$

$$s_1 = s_{g1}$$

Enthalpies at Various points



$$1-2 \quad s=c, \quad s_1 = s_2$$

$$s_1 = s_{\text{max}} = s_{f1} + x s_{fg1}$$

$$s_2 = s_{g2}$$

$$h_1 = h_{\text{wet}} = h_{f1} + x h_{fg1}$$

$$h_2 = h_{g2}$$

$$h_3 = h_{f2}$$

$$h_4 = h_3$$

$$h_4 = h_{f2} + x h_{fg2}$$

Refrigeration Capacity (R.C)

$$R.C = R.E \times m_R$$

$$R.C = q_a \times m_R$$

$$R.C = (h_1 - h_4) \cdot m_R \quad \text{kw}$$

$$\frac{\text{kJ}}{\text{kg}} \times \frac{\text{kg}}{\text{s}}$$

Power input to compressor

$$W_c = h_2 - h_1 \quad \frac{\text{kJ}}{\text{kg}}$$

$$W_c = W_c \times m_R$$

$$W_c = (h_2 - h_1) m_R$$

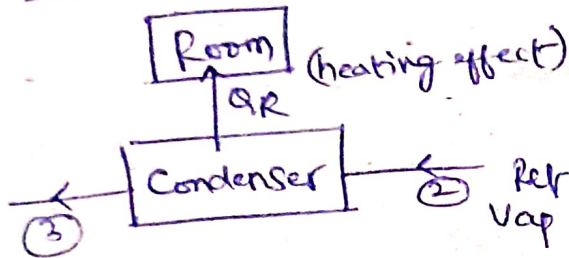
kw

$$\frac{\text{kJ}}{\text{kg}} \times \frac{\text{kg}}{\text{s}}$$

$$m_R = \frac{\text{mass flow rate of refrigerant}}{\text{kg/s}}$$

$$\text{COP}_{\text{VCR}} = \frac{\text{Ref. eff.}}{\text{Comp. Work/Kg}} = \frac{R \cdot E_{\text{mix}}}{W_{\text{C}} \cdot x_{\text{mix}} R} = \frac{\text{Ref. Cap}}{\text{Power I/P}} = \frac{R_c}{W_c} = \text{COP}_{\text{VCR}}$$

Cop of Vap Comp Heat pump

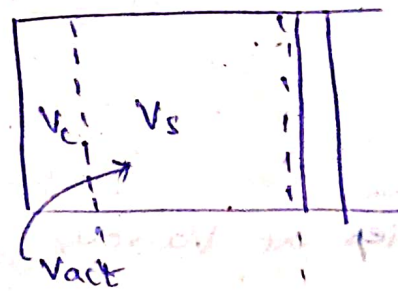


$$(\text{COP}_{\text{VCR}})_{\text{HP}} = \frac{\text{Heating effect}}{W_{\text{C}}} \\ \text{COP}_{\text{VCR HP}} = \frac{Q_R}{W_c} = \frac{h_2 - h_3}{h_2 - h_1}$$

Volumetric η of Reciprocating Comp

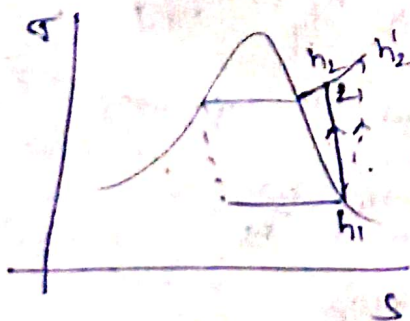
$$\eta_{\text{vol}} = \frac{V_{\text{act}}}{V_{\text{th}}} = \frac{V_{\text{act}}}{V_s} = \frac{(V_1 - V_4)}{(V_1 - V_3)}$$

$$\eta_{\text{vol}} = 1 + C \left(\frac{P_2}{P_1} \right)^{\frac{1}{n}}$$



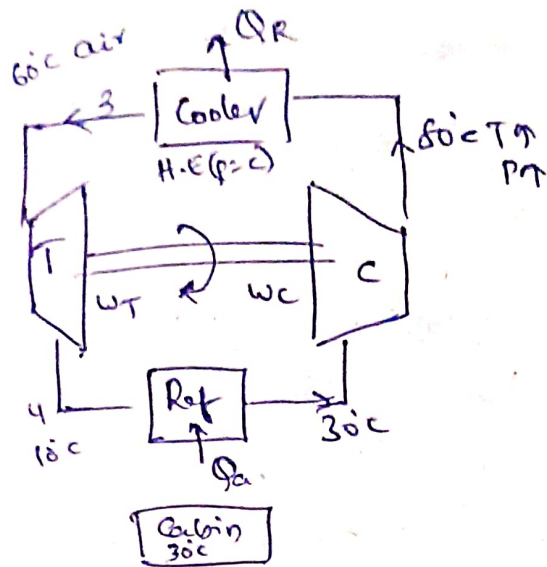
$$C = \frac{V_c}{V_s}$$

Isentropic η of Compressor

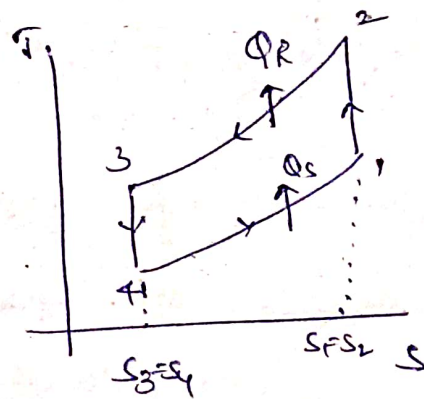
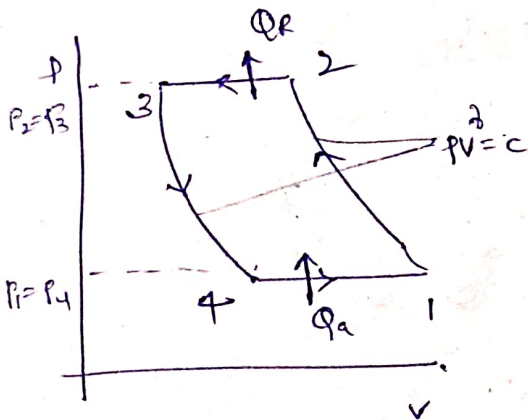


$$\eta_{\text{ise-c}} = \frac{h_{\text{ideal}}}{h_{\text{act}}} = \frac{h_2 - h_1}{h_{2'} - h_1}$$

gas Ref (or) Air Refrigeration - used in air crafts $\rightarrow$ weight is less



Rev Brayton cycle (or) Rev Joule cycle



Pressure ratio $\alpha_p = \frac{P_2}{P_1} = \frac{P_3}{P_4}$

Air Std COP of Air ref cycle

$$COP = \frac{RE}{W_{in}} = \frac{Q_a}{W_C - W_T}$$

$$= \frac{h_1 - h_4}{(h_2 - h_3) - (h_3 - h_4)} = \frac{C_p(T_1 - T_4)}{C_p[(T_2 - T_1) - (T_3 - T_4)]}$$

$$W_C = h_2 - h_1$$

$$Q_R = h_2 - h_3$$

$$Q_a = h_1 - h_4$$

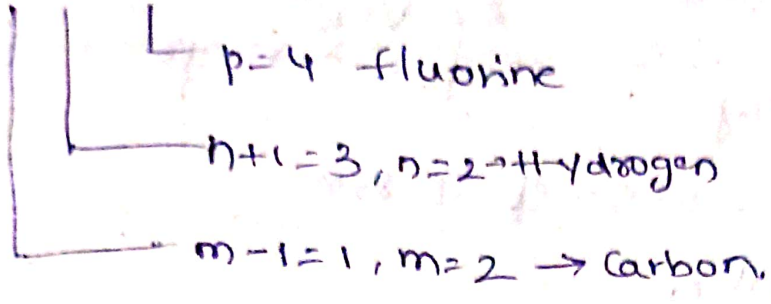
$$W_T = h_3 - h_4$$

Ideal gas: $h = c_p T$

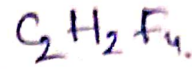
$$COP_{air R} = \frac{1}{\left[\alpha_p^{\frac{\gamma-1}{\gamma}} - 1 \right]}$$

02/10/24

R 1 3 4 a



Formula $C_m H_n F_p Cl_q$



Joule-Thomson - Coefficient

$$\mu = \left(\frac{\partial T}{\partial p} \right)_h$$

$\partial p = -ve$ always,

change in fluid temp as it flow from high pressure region to low pr region.

If $\mu = +ve$ means $\partial T = +ve$, temp drops during throttling

Summary

Relative humidity $(\phi) = \frac{P_v}{P_{vs}}$

$\xrightarrow{\text{water}}$ Vapour pressure
 $\xrightarrow{\text{sat water}}$ pressure

$P_t = P_{at} + P_v$
 $\downarrow$ atm & total pr
 $\downarrow$ air pr
 $\downarrow$ vap pr

Sp. Humidity $(w) = 0.622 \left(\frac{P_v}{P_t - P_v} \right)$

$\xrightarrow{\text{atm pressure}}$
 humidity ratio

$w = \frac{m_v}{m_a}$
 $\xrightarrow{\text{mass of vapour}}$
 $\xrightarrow{\text{mass of air}}$

Saturated Cond

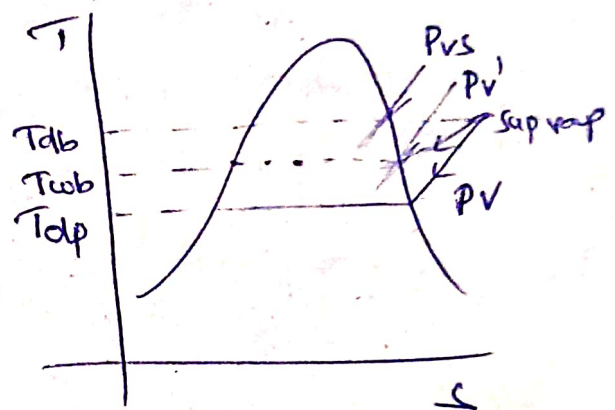
$\phi = 100\% = 1$

$\phi = \frac{P_v}{P_{vs}} = 1$

$P_v = P_{vs}$
 $\downarrow$
 $T_{dp} = T_{db}$

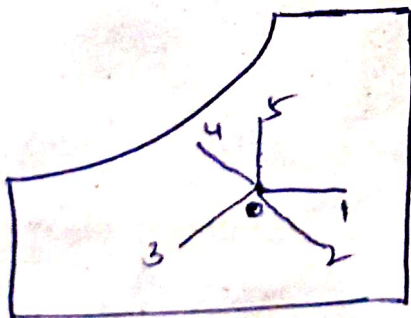
wet bulb depression $= T_{db} - T_{wb}$

$T_{db} \rightarrow P_{vs}$
 $T_{wb} \rightarrow P_v'$
 $T_{dp} \rightarrow P_v$



Dew point: DPT for the temp where water vapour condenses.

(Ex: Cool drink kept outside ~~water vapour~~ ~~find~~ Condense and form on the cool drink.)



0-5 - Humidification

0-1 - Sensible heating

0-2 = heating & dehumidification

0-3 Cooling & dehumidification

0-4 Cooling & humidification

Conservation of momentum

Mach Number (M)

$$M = \frac{v}{a}$$

v = velocity of fluid (m/s)

a = velocity of sound (m/s)

Various flows

1. Subsonic flow ($M < 1$)
2. Sonic flow ($M = 1$)
3. Transonic flow ($M \approx 1$)
4. Supersonic flow ($M > 1$)
5. Hypersonic flow ($M \gg 1$)

Velocity of Sound for an Ideal gas

$$a = \sqrt{\gamma R T}$$

m/s

$$= \sqrt{\frac{J}{kg \cdot K} \times K}$$

$$= \sqrt{\left(\frac{m}{s^2}\right)}$$

$$a = m/s$$

$\gamma = \text{No units}$

$$\frac{m^2}{s^2} = \frac{J}{kg}$$

Summary

Velocity of sound for an Ideal gas $a = \sqrt{\gamma RT}$ m/s $\left. \begin{array}{l} \\ \end{array} \right\} V_{imp}$

stagnation Enthalpy: $h_0 = h + \frac{V^2}{2}$

stagnation Temp: $T_0 = T + \frac{V^2}{2C_p}$

stagnation pressure: $P_0 = P \left(\frac{T_0}{T} \right)^{\frac{\gamma}{\gamma-1}}$

Ratio of Stagnation temp to the static temp $\frac{T_0}{T} = 1 + \left(\frac{\gamma-1}{2} \right) M^2$

Ratio of stagnation pr to static pressure $\frac{P_0}{P} = \left(1 + \left(\frac{\gamma-1}{2} \right) M^2 \right)^{\frac{\gamma}{\gamma-1}}$

Ratio of stagnation temp to critical temp $\frac{T_0}{T^*} = \frac{\gamma+1}{2}$

Ratio of stagnation pr to critical pressure $\frac{P_0}{P^*} = \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma}{\gamma-1}}$

mass flow rate in conv and divergent Nozzle

$$\frac{\dot{m}}{A} = \frac{M P_0}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R}} \left[1 + \left(\frac{\gamma-1}{2} \right) M^2 \right]^{\frac{-\gamma(\gamma+1)}{2(\gamma-1)}}$$

$$M=1$$

$$\gamma=1.4$$

$$R=287 \left(\frac{J}{kg \cdot K} \right)$$

$$\frac{\dot{m}}{A} = 0.0404 \frac{P_0}{\sqrt{T_0}}$$

- * Sign *
- Subsonic flow $M < 1$
 - Sonic flow $M = 1$
 - Transonic flow $M \approx 1$
 - Supersonic flow $M > 1$
 - Hypersonic flow $M \gg 1$

At critical portion the $M=1$ $\frac{P^*}{T^*}$

At γ

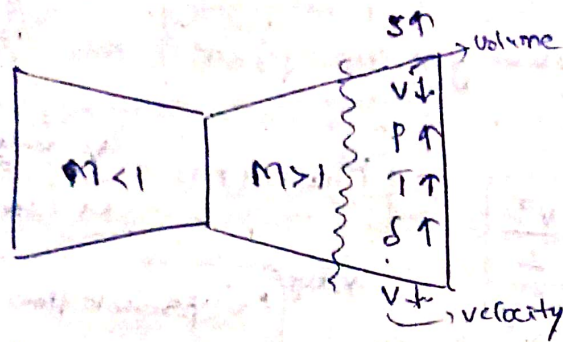
$$\gamma = 1.4$$

$$C_p = 1.005$$

$$C_v = 0.718$$

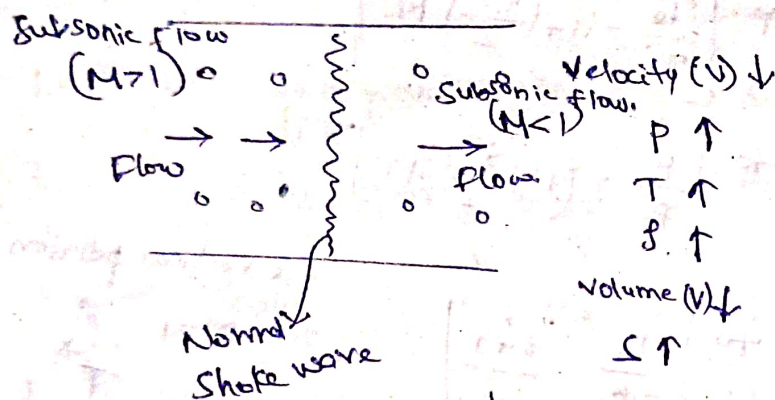
$$R = 0.287$$

always sub
this value in $J/kg \cdot K$
only



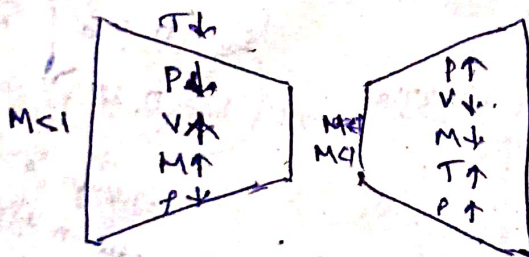
* Shockwaves $\rightarrow$ In divergent portion only is produced.

Normal Shockwave

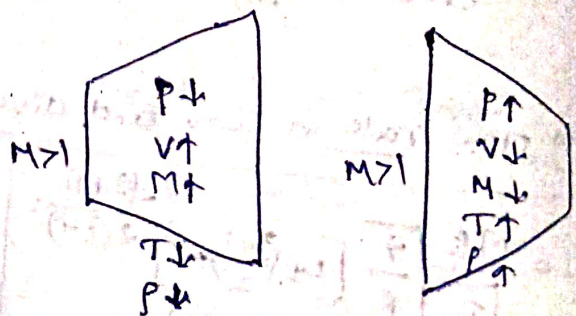


$$M = \frac{V}{a}$$

$$\uparrow \rho = \frac{m}{V} \downarrow \Rightarrow \uparrow P = \rho R T \uparrow$$



Subsonic flow



Supersonic flow

Maxwell relation

Vertically,

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_U$$

$$\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$$

Horizontally

$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

$$\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

T P

V S

TD's Equations

$$TdS = du + pdv$$

$$TdS = C_V dT + T\left(\frac{\partial P}{\partial T}\right)_V dv$$

$$TdS = dh - vdp$$

$$TdS = C_P dT - T\left(\frac{\partial V}{\partial T}\right)_P dp$$

To balance
units

↓

remaining
term so
put T